Polymer Materials

- **Polymer**
- **Monomer**
- **Oligomer**
- **Addition polymer vs condensation polymer**
- **Step-growth polymerization vs chain polymerization**
- **Thermoplastic and thermoset**
- **Configuration vs conformation**
- **Tacticity: atactic; isotactic; syndiotactic**

1. Molecular weight

Number-average molecular weight

$$\overline{M_n} = \frac{\sum N_x M_x}{N} = \frac{\sum N_x M_x}{\sum N_x} = \sum \underline{N_x}\, M_x$$

Weight-average molecular weight

$$\overline{M_w} = \sum w_x M_x = \frac{\sum N_x {M_x}^2}{\sum N_x M_x}$$

Polydispersity index (PDI)

$$PDI = M_w / M_n$$

➤ Number-average molecular weight:

$$\overline{M_n} = \frac{w}{\sum N_x} = \frac{\sum N_x M_x}{\sum N_x} = \frac{\sum N_x M_x}{N_1 + N_2 \cdots\cdots + N_x} = \frac{\sum N_x M_x}{N}$$

$$\overline{M_n} = \sum \frac{N_x}{N} M_x = \sum \underline{N_x} M_x$$

$\overline{M_n}$: number-average molecular weight;

w: total mass of the polymer;

N_x : the number of moles of molecules having a molecular weight of M_x

$\underline{N_x}$:the mole fraction (or the number-fraction) of molecules of size M_x

➢ Weight-average molecular weight:

$$\overline{M_w} = \sum w_x M_x \qquad w_x = \frac{c_x}{c}$$

$$\overline{M_w} = \frac{\sum c_x M_x}{c}$$

w_x : the weight fraction of molecules whose weight is M_x;

c_x: the weight of M_x molecules;

c: the total weight of all the polymer molecules.

$$c_x = N_x M_x$$

$$c = \sum c_x = \sum N_x M_x$$

$$\Rightarrow \quad \boxed{\overline{M_w} = \frac{\sum c_x M_x}{c} = \frac{\sum N_x {M_x}^2}{\sum N_x M_x}}$$

Question: A polymer had been mixed with 10mol with a molecular weight of 10^4 , and 5mol with a molecular weight of 10^5 , calculate their average molecular weight (the a is 0.6).

$$\overline{M_n} = \frac{\sum NiMi}{\sum Ni} = \frac{10\times10^4+5\times10^5}{10+5} = 40000$$

$$\overline{M_w} = \frac{\sum NiMi^2}{\sum NiMi} = \frac{10\times(10^4)^2+5\times(10^5)^2}{10\times10^4+5\times10^5} = 85000$$

$$\overline{M_v} = \left(\frac{10\times(10^4)^{0.6+1}+5\times(10^5)^{0.6+1}}{10\times10^4+5\times10^5}\right)^{1/0.6} \approx 80000$$

Calculates the number-average, weight-average molecular weight and PDI according to the following data.

Components	Weight Fractions	Average relative molecular weight
A	0.05	13000
B	0.09	20000
C	0.20	34000
D	0.18	45000
E	0.21	70000
F	0.13	112000
G	0.10	132000
H	0.04	156000

Assumption: the total mass of the polymer is 100 g

$$\overline{M_w} = \sum w_x M_x$$

$$\overline{M_n} = \sum \frac{N_x}{N} M_x$$

M_x	w_x	$w_x \times M_x$	c_x/g	$N_x = c_x/M_x$	$N_x/\sum N_x$	$N_x \cdot M_x/\sum N_x$
13000	0.05	650	5	0.000384615	0.164343004	2136.459058
20000	0.09	1800	9	0.00045	0.192281315	3845.626305
34000	0.2	6800	20	0.000588235	0.251348125	8545.836234
45000	0.18	8100	18	0.0004	0.170916725	7691.25261
70000	0.21	14700	21	0.0003	0.128187544	8973.128045
112000	0.13	14560	13	0.000116071	0.049596371	5554.793552
132000	0.1	13200	10	$7.57576E-05$	0.032370592	4272.918117
156000	0.04	6240	4	$2.5641E-05$	0.0109562	1709.167247

Total 66050

Total 42729.2

$$PDI = \frac{M_w}{M_n} \approx 1.55$$

Calculate number average molecular weight, weight average molecular weight and the polydispersity index (PDI) for a polymer sample that contains the following mixture.

	Mole fraction	Molecular weight
1	0.1	1000000
2	0.2	700000
3	0.4	500000
4	0.2	250000
5	0.1	100000

Assumption: the total N of the polymer is 1 mol

$$\overline{M_n} = \sum \underline{N_x} M_x$$

$$\overline{M_w} = \sum w_x M_x$$

M_x	$\underline{N_x}$	N_x	$N_x \times M_x$	$c_x = N_x \times M_x$	$w_x = \frac{c_x}{c}$	$w_x M_x$
1000000	0.1	0.1	100000	100000	0.2	200000
700000	0.2	0.2	140000	140000	0.28	196000
500000	0.4	0.4	200000	200000	0.4	200000
250000	0.2	0.2	50000	50000	0.1	25000
100000	0.1	0.1	10000	10000	0.02	2000

Total 500000

Total 623000

$$PDI = \frac{M_w}{M_n} \approx 1.246$$

2. Step-growth polymerization and Carothers' equation

p: the fraction of functional group conversion
官能团的反应率

$$p = \frac{[N_a]_0 - [N_a]}{[N_a]_0}$$

$[N_a]_0$ *is the initial* $(at\ t\ = 0)$ *concentration of the functional group* a.

$[N_a]$ *is the concentration of the functional group* a *at time* t.

$\overline{X_n}$: 聚合物的聚合度 The number-average degree of polymerization

$\overline{X_n}$: 每条高分子链所含结构单元数的平均值
Average number of structural units contained in a polymer chain

$$\overline{X_n} = \frac{\text{结构单元总数}}{\text{大分子个数}} = \frac{\text{the total number of structural units}}{\text{the number of polymer chains}}$$

2-2 体系的聚合物a[A-B]$_n$b由两种结构单元（A、B）组成重复单元（A-B）
结构单元数是重复单元数的两倍，因此$\overline{X_n}$ =2n

Functionalities = 2

t =0 aAa : bBb =1:1

$[N_a]_0$ *is the initial (at $t = 0$) concentration of the functional group a.*

能够组成结构单元的个数为$[N_a]_0$

t =t $a[A\text{-}B]_n b$

$a[A\text{-}B\text{-}A]_n a$

$b[B\text{-}A\text{-}B]_n b$

aAa

bBb

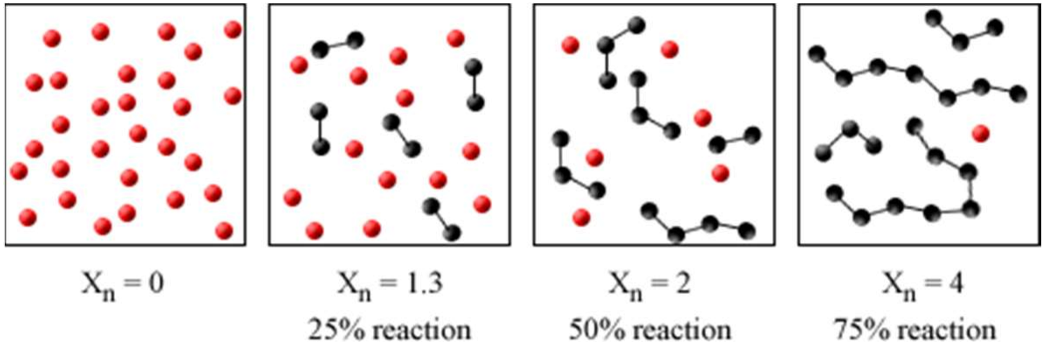
$[N_a]$ *is the concentration of the functional group a at time t.*

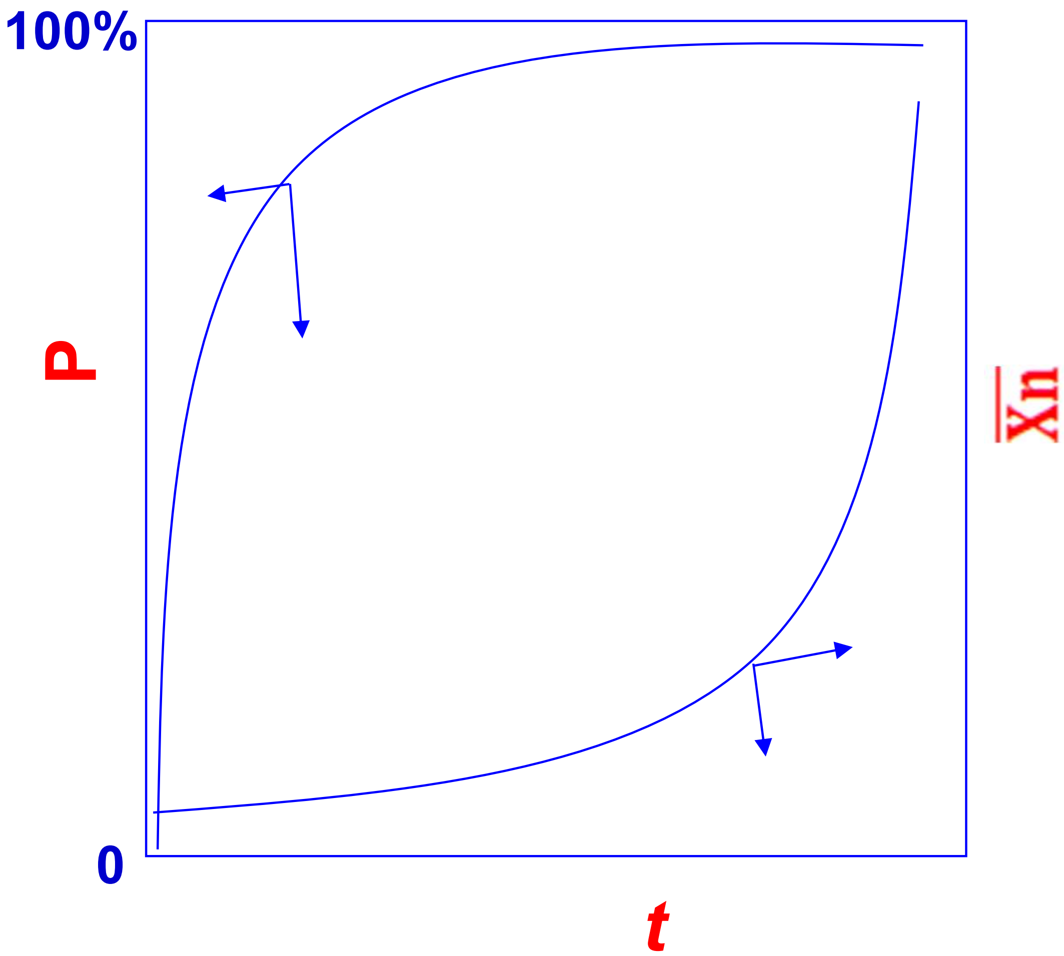
大分子的个数为N_a

$$\overline{X_n} = \frac{\text{结构单元总数}}{\text{大分子个数}} = \frac{[N_a]_0}{[N_a]}$$

$$\because p = \frac{[N_a]_0 - [N_a]}{[N_a]_0}$$

$$\therefore \overline{X_n} = \frac{1}{1-p}$$





***P*=0.9 ,$\overline{Xn}$ =10**

***p*=0.995, $\overline{Xn}$ =200**

As the reaction progresses, higher molecular weight species are formed but the degree of polymerisation increases very slowly. At 75% conversion the average $\overline{X_n}$ is still very low ($\overline{X_n}$ = 4) and very high conversion rates are required in order to obtain high molecular weights.

r: 官能团的化学计量比 $r = \frac{[N_a]_0}{[N_b]_0} < 1$

$[N_a]_0, [N_b]_0$: *the numbers of functional group a and b*

$$\overline{X_n} = \frac{\text{结构单元总数}}{\text{大分子个数}} = \frac{[N_a]_0}{[N_a]} \Rightarrow$$ 变为求平均值

结构单元总数=$\{[N_a]_0 + [N_b]_0\}/2 = \{[N_a]_0 + [N_b]_0/r\}/2 = [N_a]_0(1 + {}^1/_r)/2$

大分子个数=$\{N_a + N_b\} = \{[N_a]_0(1-p) + [N_b]_0(1-rp)\}/2$

$$\overline{X_n} = \frac{[N_a]_0(1+{}^1/_r)/2}{\{[N_a]_0(1-p)+[N_b]_0(1-rp)\}/2} = \frac{1+r}{1+r-2rp}$$

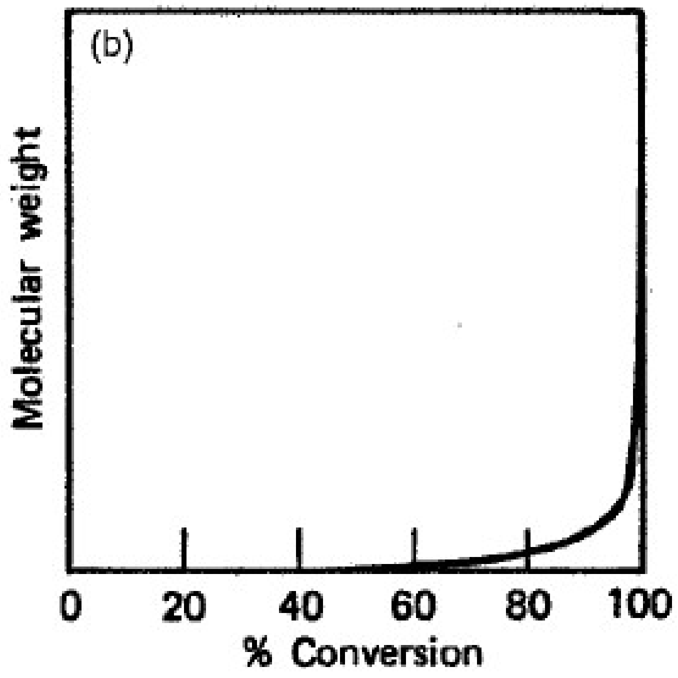
$$\overline{X_n} = \frac{1+r}{1+r-2rp}$$

Carothers' equation

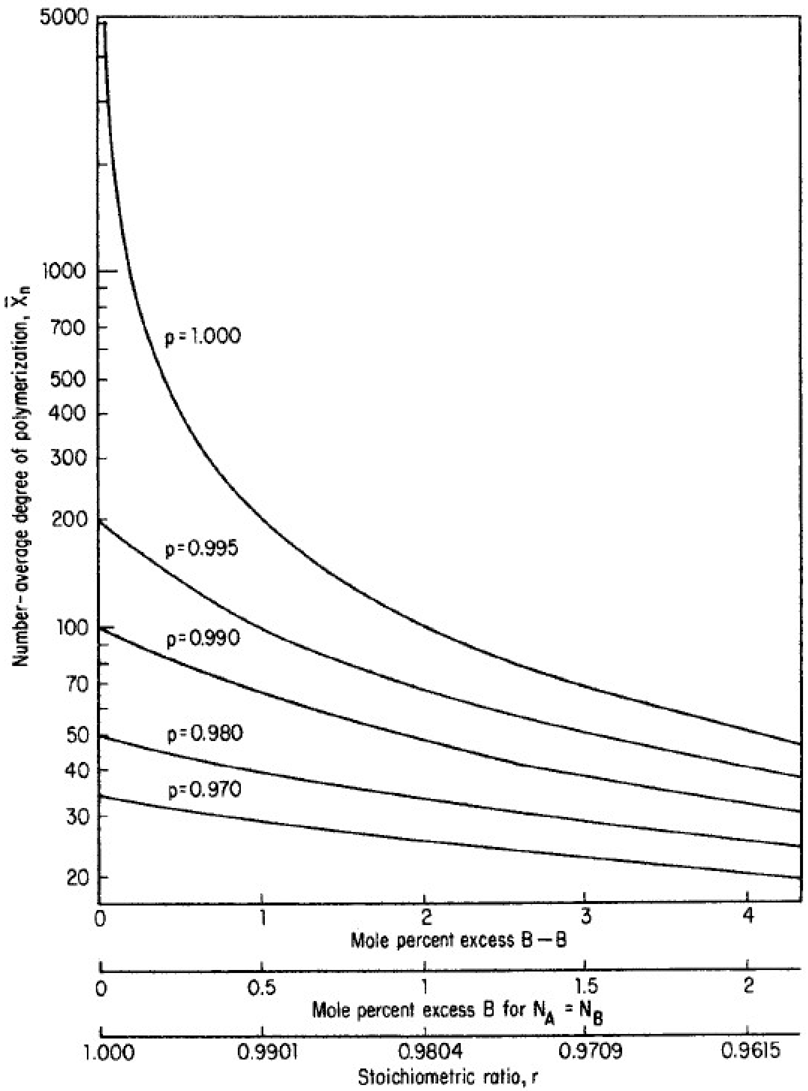
If r=1

The two bifunctional monomers are present in stoichiometric amounts

$$\overline{X_n} = \frac{1}{1-p}$$



If r < 1



If $r = \frac{1000}{1001}$ $p = 100\%$

$$\overline{X_n} = 2001$$

If $r = \frac{100}{101}$ $p = 100\%$

$$\overline{X_n} = 201$$

$r \downarrow$

$\overline{X_n}$ *decrease significantly*

It is clear that step polymerizations will almost always be carried out to at least 98% reaction, since a degree of polymerization of at least approximately 50–100 is usually required for a useful polymer. Higher conversions and the appropriate stoichiometric ratio are required to obtain higher degrees of polymerization.

Functionalities > 2

stoichiometric Reactant Mixtures

- The average functionality (f_{avg})

$$\boxed{f_{avg} = \frac{\sum N_i f_i}{\sum N_i}}$$

N_i : the reacting functionality f_i of monomer i,

$\sum N_i$: the summations are over all the monomers present in the system.

$$p = \frac{2(N_0 - N)}{N_0 f_{avg}} \qquad \overline{X}_n = \frac{N_0}{N} \quad \Rightarrow \quad \overline{X}_n = \frac{2}{2 - p f_{avg}}$$

$$p = \frac{2}{f_{avg}} - \frac{2}{\overline{X}_n f_{avg}} \quad \xrightarrow{\overline{X}_n \to \infty} \quad \boxed{p_c = \frac{2}{f_{avg}}}$$

$$\boxed{f_{avg} = \frac{\sum N_i f_i}{\sum N_i}}$$

N_0: the number of monomer molecules present initially;

$N_0 f_{avg}$: the total number of functional groups;

N : the number of molecules after reaction has occurred;

$2(N_0 - N)$: the number of functional groups that have reacted.

Functionalities>2 & Nonstoichiometric Reactant Mixtures

Ternary mixture: N_A moles of A_{f_A}, N_C moles of A_{f_C} $(fc > 2)$, NB moles of B_{f_B}. A and C have same functional groups (such as A); Number of A functional group is smaller than that of B functional group.(i.e. B groups are in excess)

$$f_{avg} = \frac{2(N_A f_A + N_C f_C)}{N_A + N_C + N_B}$$ (Only calculate the reacting groups)

The extent of polymerization depends on the deficient reactant. The excess of other reactant results in lowering of functionality.

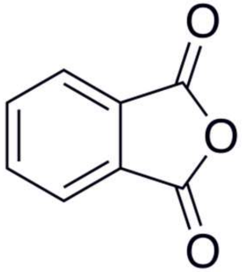
Calculate the extent of reaction at which gelation occurs for the following mixtures:

a. Phthalic anhydride and glycerol in stoichiometric amounts

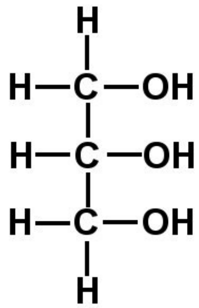
b. Phthalic anhydride and glycerol in the molar ratio 1.500:0.980

c. Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.990:0.02

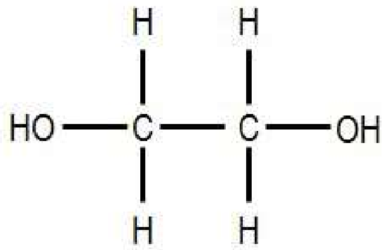
d. Phthalic anhydride, glycerol and ethylene glycol in the molar ratio 1.500:0.500:0.700



$f_B = 2$



$f_C = 3$



$f_A = 2$

<table>
<tr><th></th><th>$N_B f_B$</th><th>$N_C f_C$</th><th>$N_A f_A$</th></tr>
<tr><td>a</td><td>$3 \times 2 = 6$</td><td>$2 \times 3 = 6$</td><td></td></tr>
<tr><td>b</td><td>$1.5 \times 2 = 3$</td><td>$0.98 \times 3 = 2.94$</td><td></td></tr>
<tr><td rowspan="2">c</td><td rowspan="2">$1.5 \times 2 = 3$</td><td>$0.99 \times 3 = 2.97$</td><td>$0.02 \times 2 = 0.04$</td></tr>
<tr><td colspan="2">3.01</td></tr>
<tr><td rowspan="2">d</td><td rowspan="2">$1.5 \times 2 = 3$</td><td>$0.500 \times 3 = 1.5$</td><td>$0.700 \times 2 = 1.4$</td></tr>
<tr><td colspan="2">2.9</td></tr>
</table>

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	$\boldsymbol{f}_{\text{avg}}$	$p_c = \dfrac{2}{f_{avg}}$
a	$\dfrac{6+6}{3+2} = 2.4$	0.83
b	$\dfrac{2.94 \times 2}{1.5 + 0.98} = 2.37$	0.84
c	$\dfrac{3 \times 2}{1.5 + 0.99 + 0.02} = 2.39$	0.84
d	$\dfrac{2.9 \times 2}{1.5 + 0.5 + 0.7} = 2.15$	0.93

3. Chain-growth polymerization and Mayo equation

- Free-radical polymerization

- Ionic polymerization
  - (Living) Anionic polymerization
  - Cationic polymerization

  Ring-open polymerization

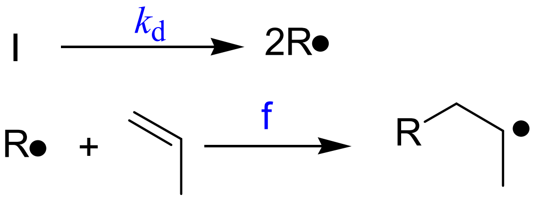
- Coordination polymerization: the only chain polymerization can control tacticity

- Living free radical polymerization: *Three important features for living polymerization*.

1. Chain termination and chain transfer reactions are absent.
2. The rate of chain initiation is much larger than the rate of chain propagation, initiators decompose quickly to insure narrow MWD
3. The polymer chains grow at a more constant rate than seen in traditional chain polymerization and their lengths remain very similar.

Four steps in free-radical polymerization

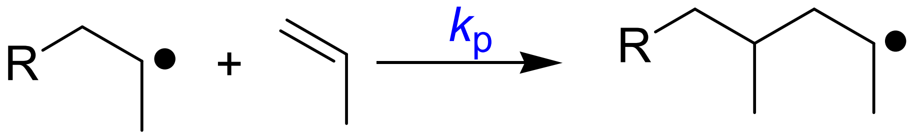
1. Initiation: homolytic cleavage(均裂) to generate radical (through heat or UV), then first addition of monomer:



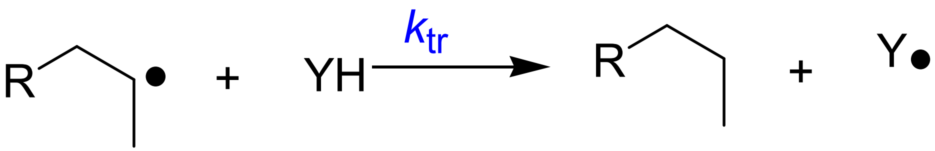
(Radical not necessarily identical)

fast (for useful initiators); f: initiator efficiency.

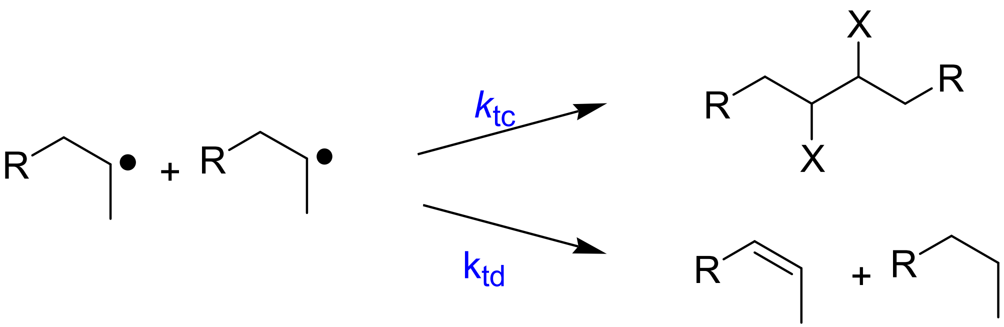
2. Propagation: repeat many times to form high-M_W polymer; "chain growth":



3. Transfer: often happens, whether desired or not:.



4. Termination-by either coupling (usually) or disproportionation (sometimes):



coupling

disproportionation

- 聚合速率和哪些因素有关

$$[M\cdot] = \text{constant} \Rightarrow R_i = R_t$$

This means that radicals are consumed at the same rate as they are generated.

$$\boxed{R_p = k_p[\frac{k_d}{k_t}]^{1/2}[fI]^{1/2}[M] = k\,[fI]^{1/2}[M]}$$

R_p: rate of polymerization

k_d: rate constant of initiation

k_p: rate constant of propagation

k_t: rate constant of termination

f: the fraction of radicals initiate chain growth

[I]: the concentration of initiator

[M]: the concentration of monomer

$$k = k_p[\frac{k_d}{k_t}]^{1/2}$$

Case 1: For a polymer, T_1 = 90℃, T_2 = 100 ℃, E = 80 kJ/mol, $R = 8.314\, J/mol \cdot K$, please calculate k_2/k_1

Arrhenius equation: $\boldsymbol{k = Ae^{-E/RT}}$

$$R_p = k_p[\frac{fk_d}{k_t}]^{1/2}[I]^{1/2}[M] = k\,[fI]^{1/2}[M]$$

$$k = Ae^{-E/RT}$$

$$\frac{k_2}{k_1} = \frac{e^{-E/RT_2}}{e^{-E/RT_1}} = e^{\frac{E}{R}(\frac{1}{T_1}-\frac{1}{T_2})} = e^{\frac{80000}{8.314}(\frac{1}{363}-\frac{1}{373})} = e^{0.71} = 2$$

Case 2: The first system: $K_2S_2O_8$ E_d = 140 kJ/mol;

second: $K_2S_2O_8$—F_e^{2+} E_d = 50 kJ/mol,

E_p = 29 kJ/mol, E_t = 17 kJ/mol, T = 60℃,

R = 8.314 kJ/mol.K, please calculate k_2/k_1。

Arrhenius equation: $k = Ae^{-E/RT}$

$$\boxed{R_p = k_p[\frac{fk_d}{k_t}]^{1/2}[I]^{1/2}[M] = k\,[fI]^{1/2}[M]}$$

$$k = k_p[\frac{k_d}{k_t}]^{1/2}$$

$$k = A_p(\frac{A_d}{A_t})^{1/2}e^{-[(E_p-E_t/2)+E_d/2]/RT}$$

$$E_1 = (E_p - \frac{E_t}{2}) + \frac{E_d}{2} = 29 - \frac{17}{2} + \frac{140}{2} = 90.5$$

$$E_2 = (E_p - \frac{E_t}{2}) + \frac{E_d}{2} = 29 - \frac{17}{2} + \frac{50}{2} = 45.5$$

$$\frac{k_2}{k_1} = e^{-(E_2-E_1)/RT} = e^{(90.5-45.5)\times1000/8.314\times333} = e^{16.25} = 1.14\times10^7$$

- 影响分子量的因素

The kinetic chain length v of a radical chain polymerization is defined as the average number of monomer molecules consumed (polymerized) per each radical, which initiates a polymer chain.

$$\nu = \frac{R_p}{R_i} = \frac{R_p}{R_t}$$

Coupling termination $\overline{X_n} = 2v$

Disproportion termination $\overline{X_n} = v$

The fractions of propagating chains, a and $(1-a)$, respectively, which undergo termination by coupling and disproportionation

$$\overline{X_n} = \frac{R_p}{(2-a)(R_i/2)}$$

只考虑了termination step

既termination又考虑chain transfer对聚合度的影响

$$\boxed{\frac{1}{\overline{X}_n} = \frac{(2-a)R_i}{2R_p} + C_{\mathrm{M}} + C_{\mathrm{S}}\frac{[\mathrm{S}]}{[\mathrm{M}]} + C_{\mathrm{I}}\frac{[\mathrm{I}]}{[\mathrm{M}]}}$$

Mayo Equation

$\underline{chain-transfer\ constant\ C}$

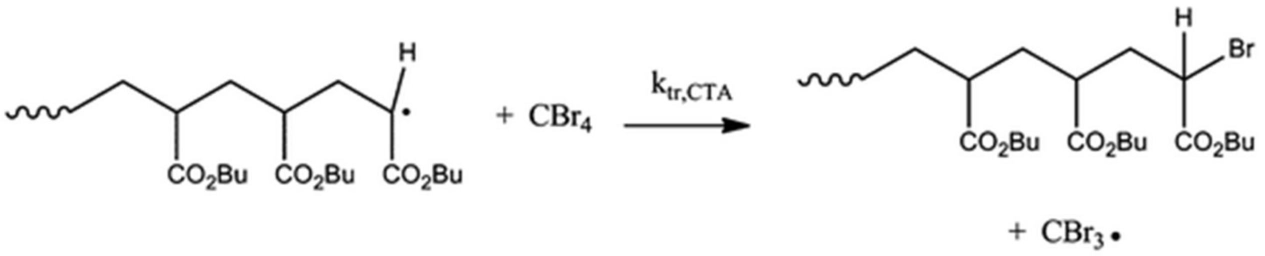
$$C_{\mathrm{M}} = \frac{k_{tr,\mathrm{M}}}{k_p} \quad C_{\mathrm{S}} = \frac{k_{tr,\mathrm{S}}}{k_p} \quad C_{\mathrm{I}} = \frac{k_{tr,\mathrm{I}}}{k_p}$$

$$\frac{1}{\overline{X}_n} = \frac{(2-a)R_i}{2R_p} + C_{\mathrm{M}} + C_{\mathrm{S}}\frac{[\mathrm{S}]}{[\mathrm{M}]} + C_{\mathrm{I}}\frac{[\mathrm{I}]}{[\mathrm{M}]}$$

Monomer transfer constants C_M

$$M_n\cdot + CH_2{=}\underset{Y}{\overset{H}{C}} \longrightarrow M_n{-}H + CH_2{=}\underset{Y}{C}\cdot$$

Initiator transfer constants C_I



Transfer to chain-transfer agent C_S

Explain the effects of the initiator concentration on the degree of polymerization (free-radical route) of styrene.

$$R_i = \frac{d[M_1 \cdot]}{dt} = 2fk_d[I]$$

A higher initiator concentration results in a higher R_i

$$\frac{1}{\overline{X}_n} = \frac{(2-a)R_i}{2R_p} + C_M + C_S\frac{[S]}{[M]} + C_I\frac{[I]}{[M]}$$

A higher R_i results in a lower degree of polymerization.

Case 1. A reaction system that yields a molecular weight lower than needed.

- The initiation rate would be lowered if the low molecular weight is due to an excessively high initiation rate.
- If the initiator or solvent are too active in transfer, a less active initiator or solvent would be chosen.
- If the monomer itself is too active in transfer, it sets the upper limit of the molecular weight that can be achieved as long as all other factors are optimized.

Case 2. A reaction system that yields too high a polymer molecular weight.

- The simplest method of lowering molecular weight is to add a chain-transfer agent. When used in this manner, transfer agents are called regulators or modifiers.

- Transfer agents with transfer constants of one or greater are especially useful, since they can be used in low concentrations.

- For example, low-molecular-weight acrylic ester polymers (聚丙烯酸酯) have been used as plasticizers.

What is the difference among free-radical polymerization, anionic /cationic polymerization, and coordination polymerization? (Please take examples to explain)

Different Initiator

Different termination

- free-radical polymerization: termination between two growing chains; chain transfer.
- Anionic polymerization: no termination and chain transfer.
- Cationic polymerization: easy chain transfer and hard termination.

free-radical polymerization vs coordination polymerization

Tacticity ⇨ Free-radical polymerization produces branched PE (LDPE)
Coordination polymerization produces linear PE (HDPE)

What is the advantage/character of living polymerization?

Characteristic of Living Polymerizations：

- Quick initiation
- Slow propagation
- No termination

Advantages:

- Superior molecular weight and polydispersity control,
- Ease of block copolymerization,
- Route to unique materials and architecture.

If I want to use living polymerization method to make a polystyrene with a molecular weight of Mn = 10400 with an assumed monomer conversion of 80%, what molar ratio between the styrene monomer and its initiator should I use?

$$\overline{X_n} = \frac{M_n}{the\ molecular\ weight\ of\ a\ repeat\ unit} = \frac{10400}{104} = 100$$

$$\overline{X_n} = \frac{p[M]_0}{[C]}$$

$[M]_0$ is the initial concentration of monomers

$[C]$ is the concentration of initiators

$$\frac{[M]_0}{[C]} = \frac{\overline{X_n}}{p} = \frac{100}{0.8} = 125$$

4. Step-growth vs Chain-growth polymerization

Examine the polymeric materials in your immediate vicinity and determine whether they were formed via step-growth polymerization or chain polymerization.

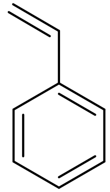
Tell the difference among step-growth polymerization, chain polymerization, condensation polymerization, and addition polymerization.

Step polymerization

- **Nylons-Polyamides**
- **PET-Polyester (polyethyleneterephthalate)- Transesterification involving a carboxylic acid ester and an alcohol**
- **PC (Polycarbonates)- Polyester formation though reaction of an acid chloride with an alcohol**

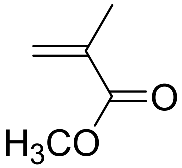
Chain polymerization-unsaturated monomer

Styrene-conjugated unsaturation; radical stabilized by π-delocation苯乙烯



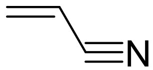
makes polystyrene, PS,

Methyl methacrylate-conjugated; works generally for acrylates and methacrylates甲基丙烯酸甲酯



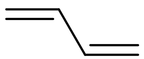
makes poly(methyl methacrylate), PMMA; Lucite(透明树脂), plexiglas（树脂玻璃）; acrylates and methacrylates are also basis of latex paints（乳胶漆）

Acrylonitrile-conjugated; typically copolymerized with styrene丙烯腈



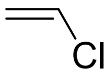
makes poly(acrylonitrile), PAN: orlon（腈纶）; acrylic fibers

1,3-butadiene-conjugated丁二烯



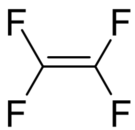
makes polybutadiens, PB: synthetic rubber

Vinyl chloride-not conjugated but polar; Cl provides radical stability氯乙烯



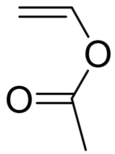
makes poly(vinyl chloride), PVC: Plastic pipes; tygon tubing

Tetrafluoroethylene-somewhat polar 四氟乙烯



makes poly(tetrafluoro ethylene), PTFE: Teflon.

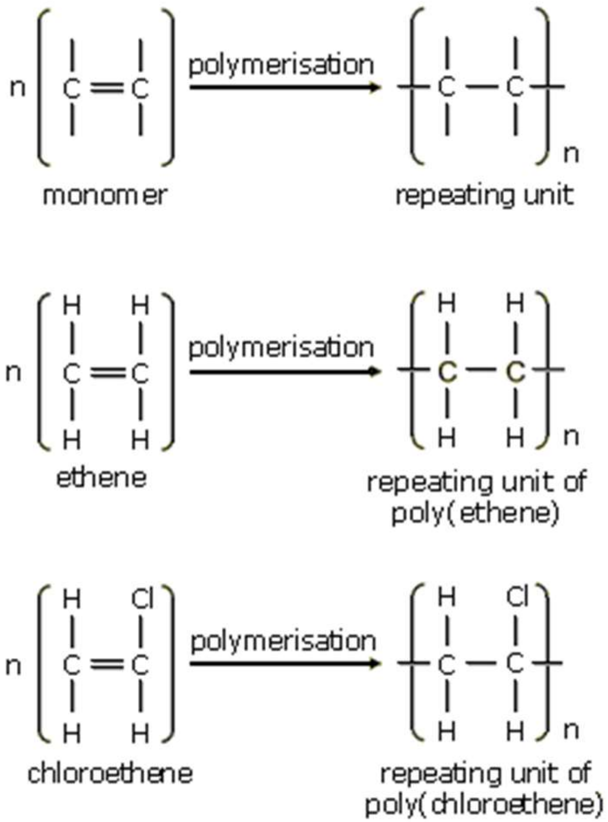
Vinyl Acetate-conjugated by ester groups.醋酸乙烯酯



makes polyvinyl alcohol.

5. Some concepts

Show the relationship between monomer and repeating unit.

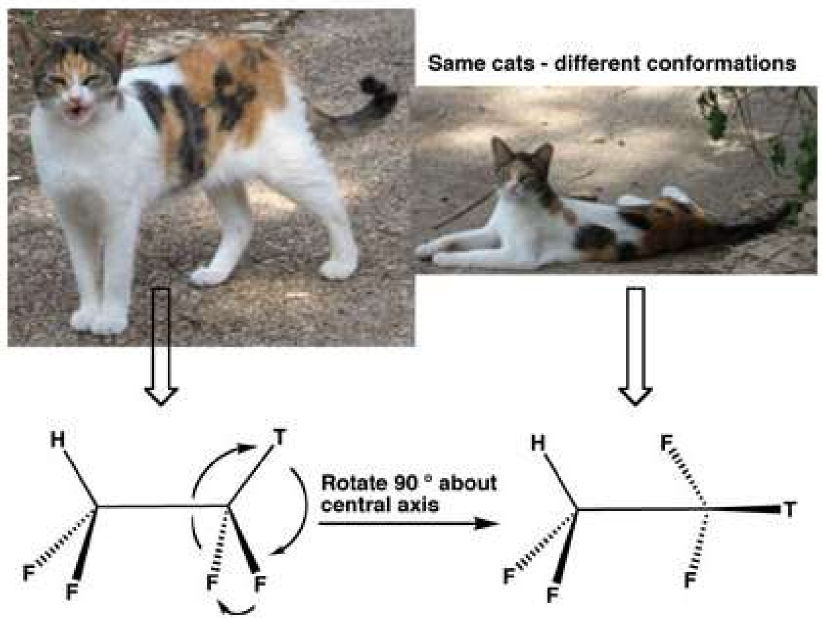


repeat unit - the fundamental recurring unit of a polymer

monomer - the smaller molecule(s) that are used to prepare a polymer (may or may not be equivalent to the repeat unit)

List a few examples for the configuration affecting polymer properties.

Different Configuration

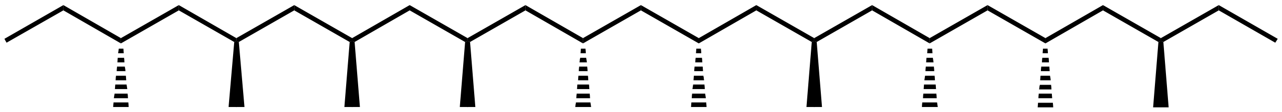


Different Conformation

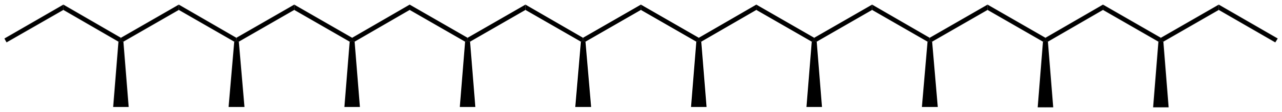
Conformations can interconvert, but **configurations** can't be interconverted without breaking off body parts and moving them around.

Polyolefins

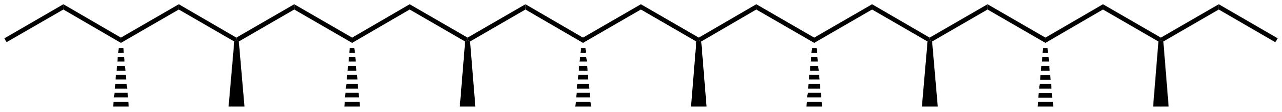
Atactic: random orientation



Isotactic: All stereocenters have same orientation



Syndiotactic: Alternating stereochemistry



- Tacticity affects physical properties

  - Atactic polymers will generally be amorphous, soft, flexible materials.

  - Isotactic and syndiotactic polymers will be more crystalline, thus harder and less flexible.

- Polypropylene (聚丙烯PP) is a good example

  - Atactic PP is a low melting, gooey(胶粘的)material.

  - Isoatactic PP is high melting (165º), crystalline, tough material that is industrially useful.

  - Syndiotactic PP has similar properties, but is very clear. It is harder to synthesize.

Low Density Polyethylene

Produced by free radical polymerization of ethylene gas

Polymer chain with branching, low crystallinity

More flexible, melting point ~ 115 °C; Density ~ 0.91-0.94

Uses:

– film and sheet-packaging

– trash bags

– household wrap, toys

– squeeze bottle, kitchenwares

– wire and cable insulation, and many others

High Density Polyethylene

Synthesized from ethylene using coordination polymerization

Linear polymer molecule with no branches, m.p. = 133-138 °C, specific gravity ~ 0.96

High **crystallinity** (70-90% vs. 40-60%), more opaque than LDPE

Stiffer, harder, higher tensile strength than LDPE

Uses

-containers and lids

-food bottles, petrol tank

-motor oil bottles

HDPE is much stronger than LDPE, but LDPE is cheaper and easier to make.

CHAPTER II

Solid-State Chemistry

1. Amorphous vs crystalline solids

Crystalline Solids	Amorphous solids
They have a definite shape and geometrical form.	They do not have a definite geometrical shape.
They have a sharp (definite) melting point.	They melt over a wide range of temperatures.
They are rigid and incompressible.	They too are usually rigid and cannot be compressed to any appreciable extent. However graphite is soft because of its unusual structure.
They give a clean clevage, i.e, break into pieces with plane surfaces.	They give irregular clevage.
They have a definite heat of fusion.	They do not have a definite heat of fusion.
Anisotropic, i.e. their mechanical properties and electrical properties depend on the direction along which they are measured.	Isotropic, i.e. they have similar physical properties in all directions because the constituents are arranged in random manner.

2. Four types of bonding in solids

Types of Solid	Units	Forces	Properties
Ionic	Positive and negative ions	Electrostatic attractions	Hard and brittle; High Tm; Poor conductivity.
Metallic	atoms	Metallic bonds	Soft to very hard; Low to very high Tm; Excellent conductivity; Malleable and ductile.
Molecular	Atoms or molecules	London dispersion, dipole-dipole, hydrogen bonding	Fairly soft; Low to moderately high Tm; Poor conductivity.
Covalent network	Atoms connected in a network of covalent bonds	Covalent bonds	Very hard; Very high Tm; Often poor conductivity.

- Ionic bonds
  - ✓ sizes of ions;
  - ✓ charges of ions;
  - ✓ the specific arrangement of ions.
- Molecular bonds are caused by molecule polarity

$$\left.\begin{array}{l}\text{London dispersion forces}\\ \text{Dipole-dipole interactions}\end{array}\right\}\ \text{van der Waals forces}$$

Hydrogen bonding

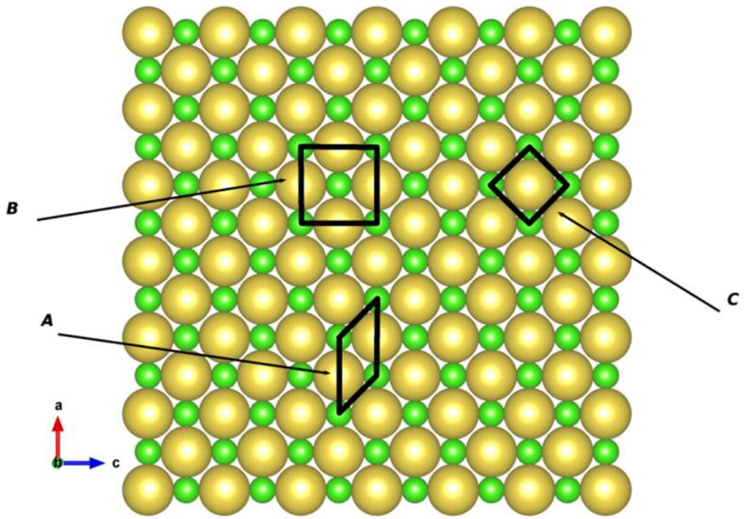
- Metallic bonding refers to the interaction between the delocalized electrons and the metal nuclei.
- Covalent network

  Carbon materials: diamond and graphite

3. Crystals

- Unit cell

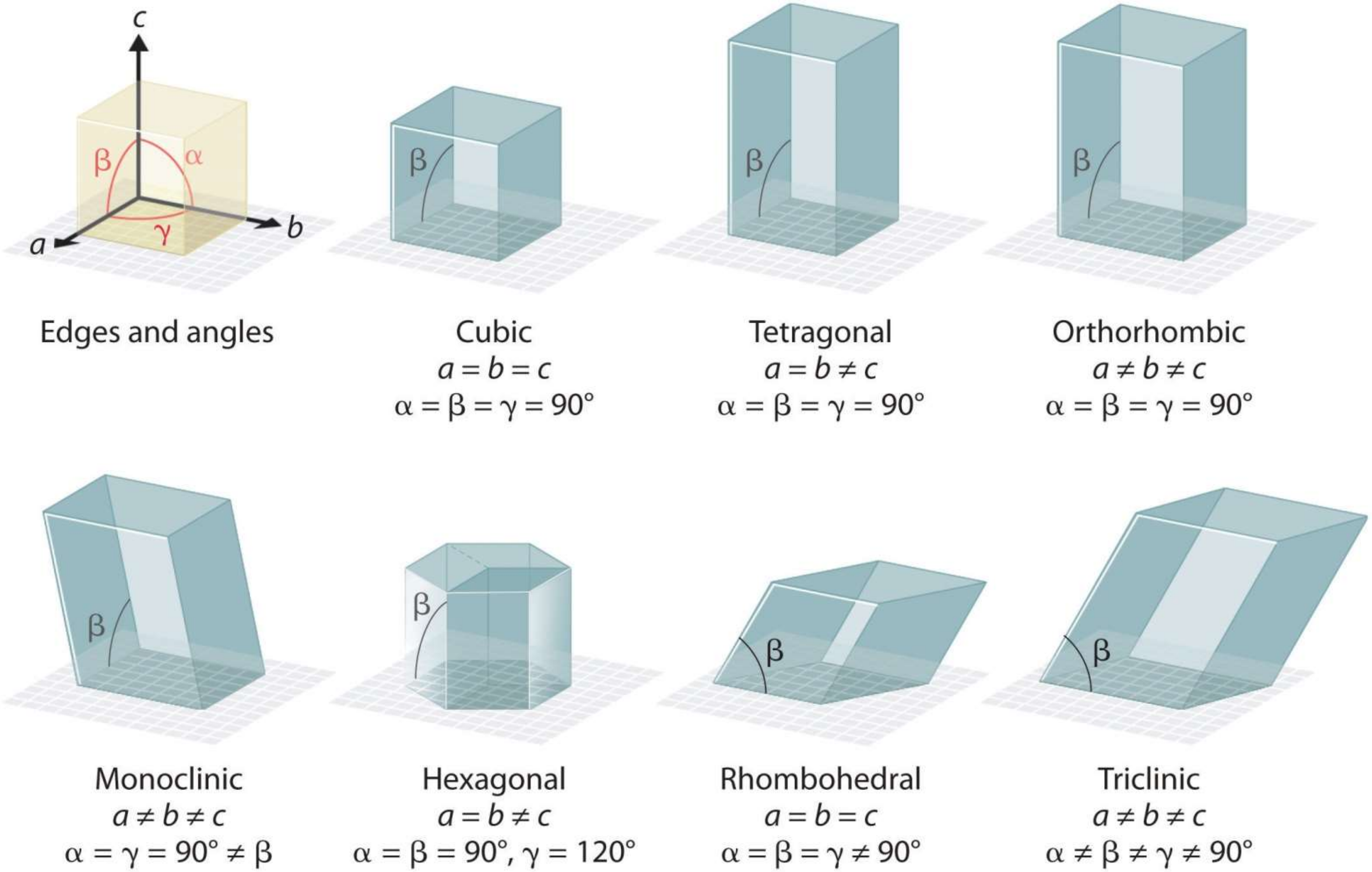
A **unit cell** is the smallest repeating unit in space lattice, which governs the symmetry and structure of the overall bulk crystal.



- **the volume of the unit cell;**
- **the symmetry of the unit cell.**

  - The unit cell is the simplest repeating unit in the crystal.
  - Opposite faces of a unit cell are parallel.
  - The edge of the unit cell connects equivalent points.

- 7 Unit cells



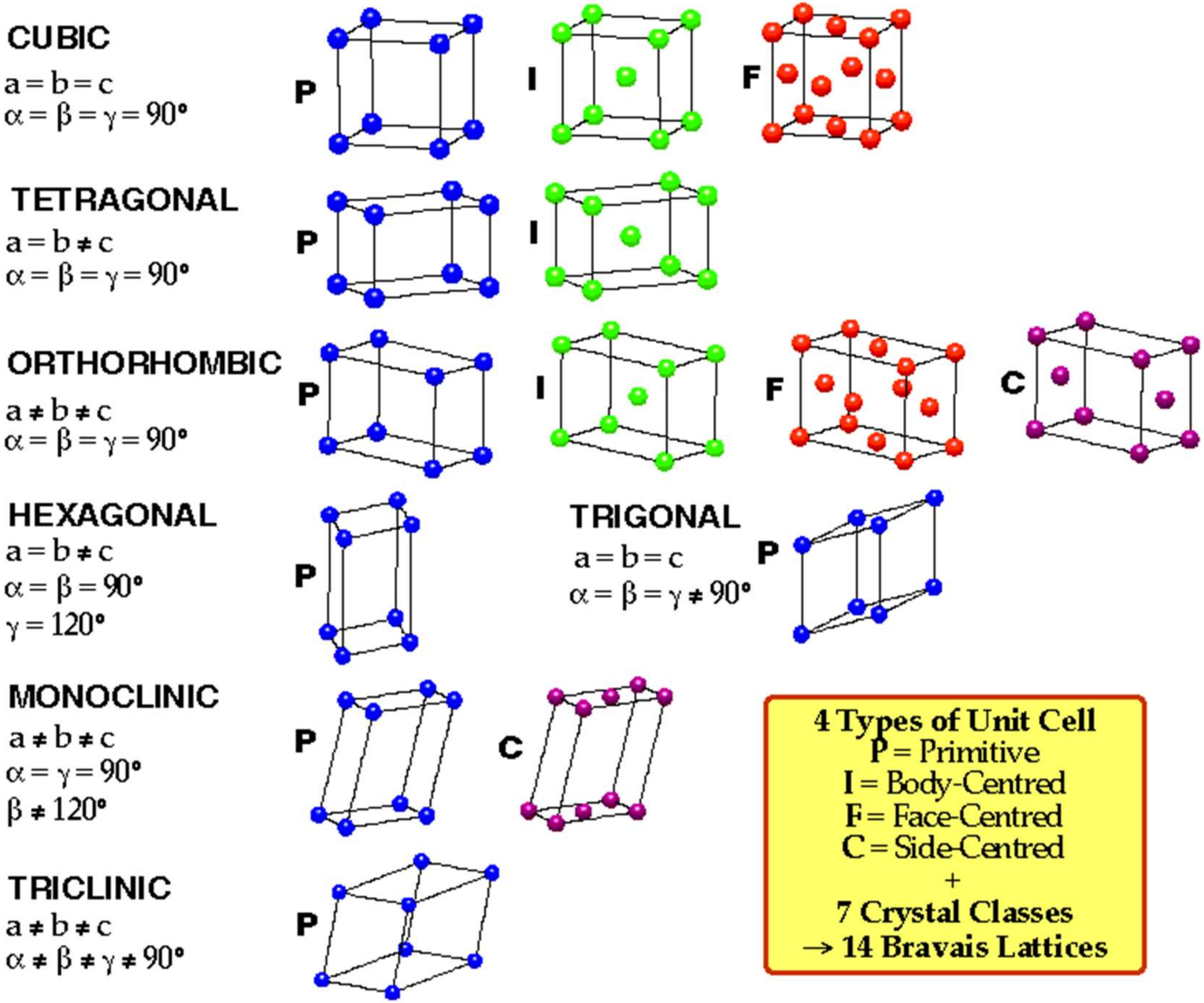
要求会匹配七大晶系和其晶格参数

Unit Cell

- **Primitive Unit Cell**: Particles are only present at corners
- **Centered Unit Cell**: Particles are present at corners and other positions
  - **Body-Centered**: Contains one particle at body center
  - **Face-Centered**: Contains one particle at the center of each face.
  - **Side-Centered**: Contains one particle at the center of any two opposing faces.

Position in unit cell	Fraction in unit cell
Center	1
Face	½
Edge	¼
Corner	⅛

- 14 Bravais unit cells

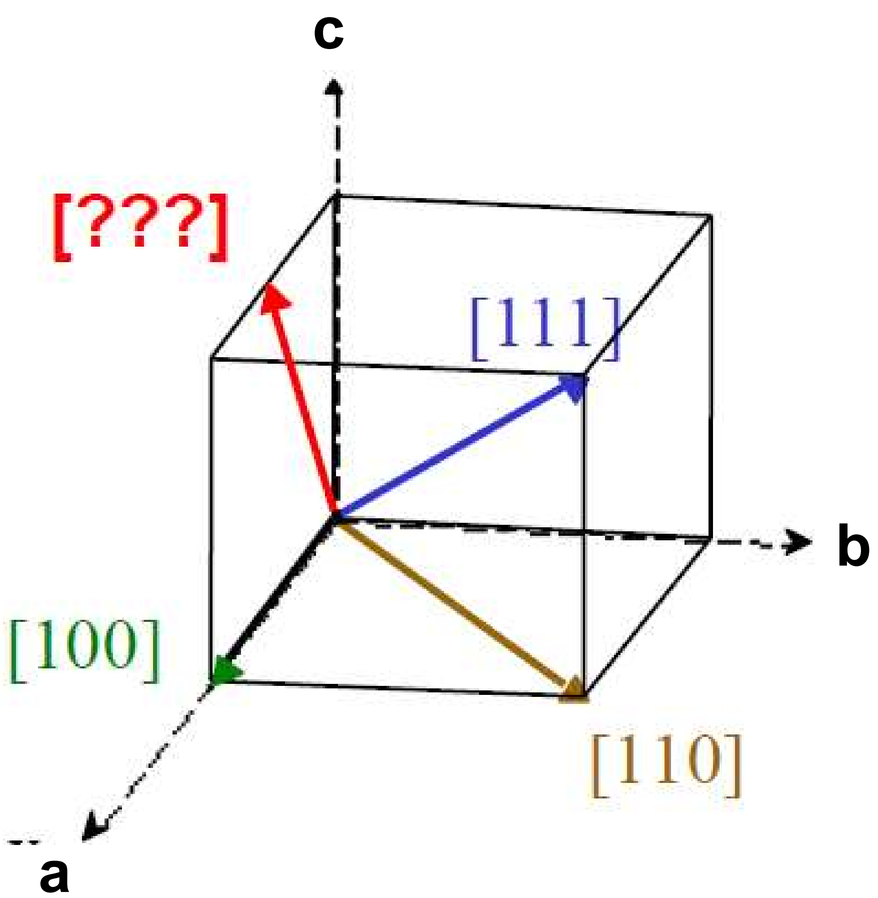


General Rules for Lattice Directions, Planes & Miller Indices

- Miller indices used to express lattice *planes* and *directions*
- a, b, c are the axes (on arbitrarily positioned origin)
- |a|, |b|, |c| are lattice parameters (length of unit cell along a side)
- h, k, l are the Miller indices for planes and directions - expressed as **planes: (hkl)** and **directions: [hkl]**

- Conventions for naming
– There are NO COMMAS between numbers
– Negative values are expressed with a bar over the number
- Example: -2 is expressed $\bar{2}$

Miller Indices for Directions



Method

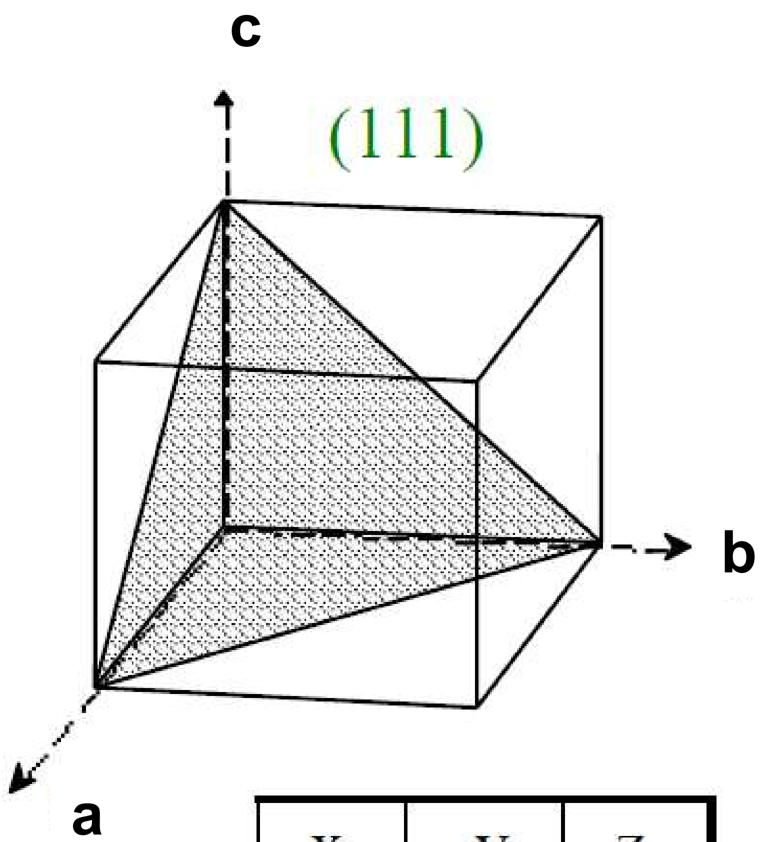
– Draw vector, and find the coordinates of the head, a_1,b_1,c_1 and the tail a_2,b_2,c_2.

– subtract coordinates of tail from coordinates of head.

– Remove fractions by multiplying by smallest possible factor.

– Enclose in square brackets.

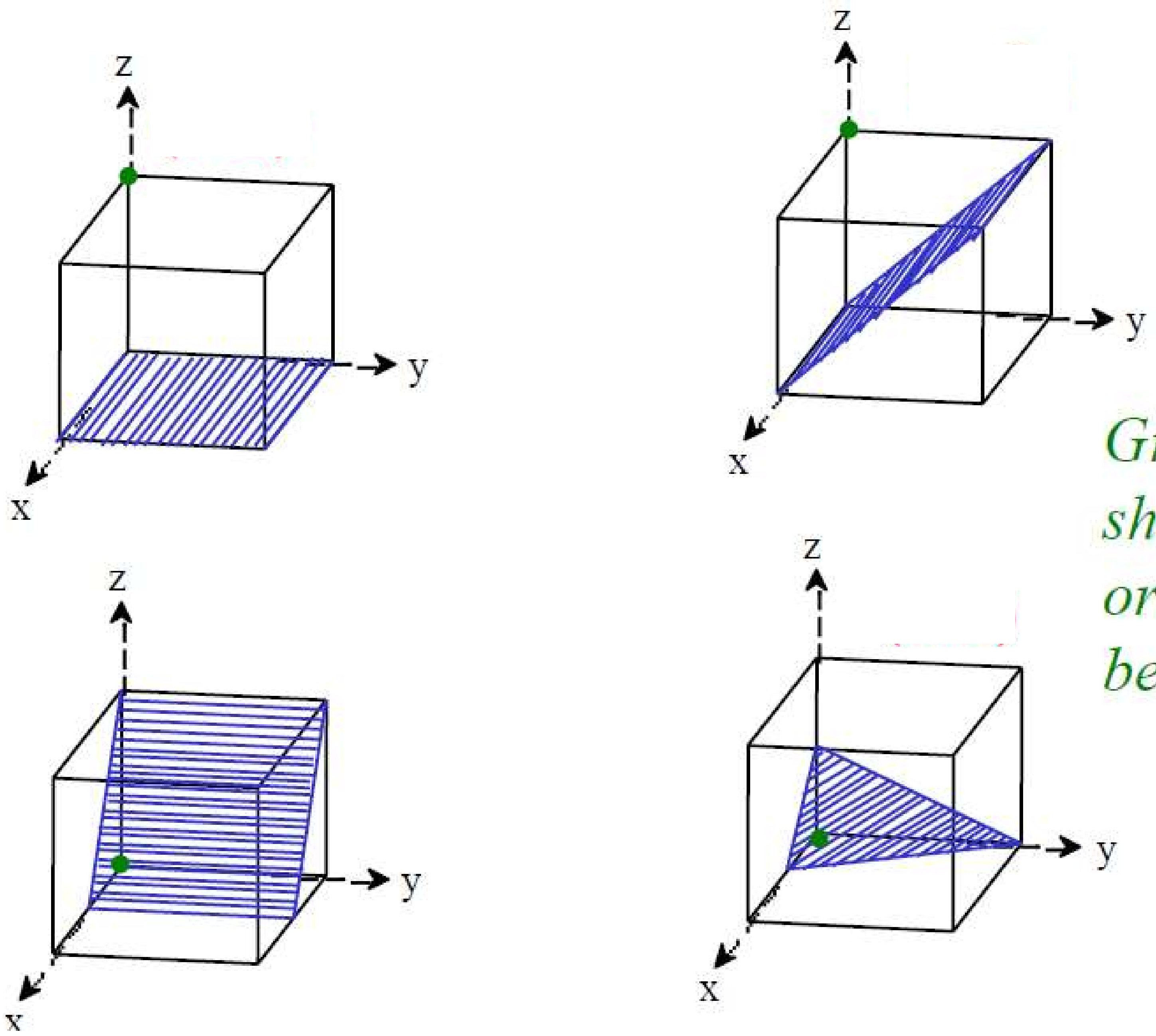
Miller Indices for Planes



	x	y	z
Intercepts	1	1	1
Reciprocals	1	1	1

Method

- If the plane passes through the origin, select an equivalent plane or move the origin.
- Determine the intersection of the plane with the axes in terms of a, b, and c
- Take the reciprocal ($1/\infty = 0$)
- Convert to smallest integers (optional)
- Enclose by parentheses

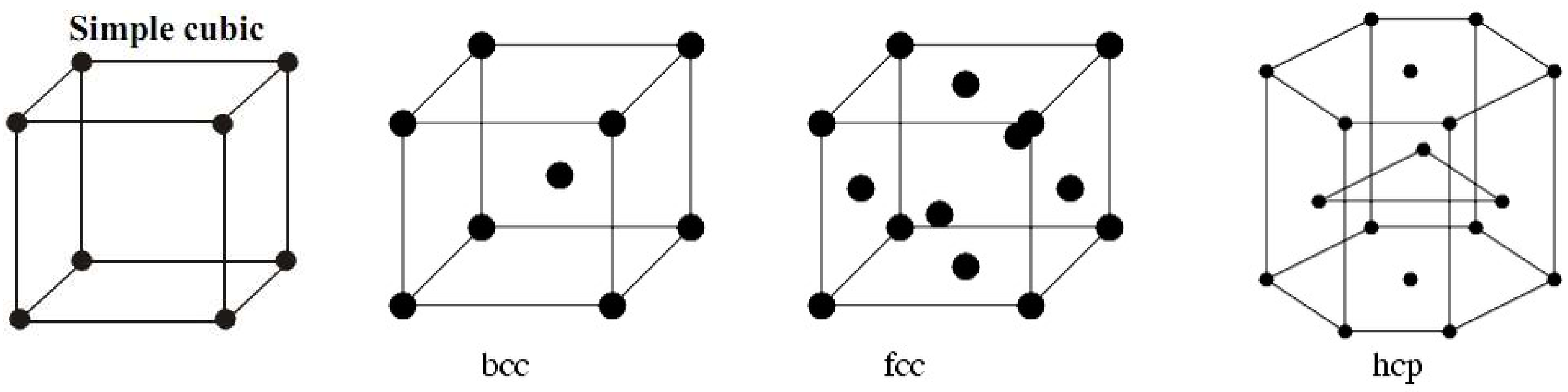


Green circles show where the origins have been placed.

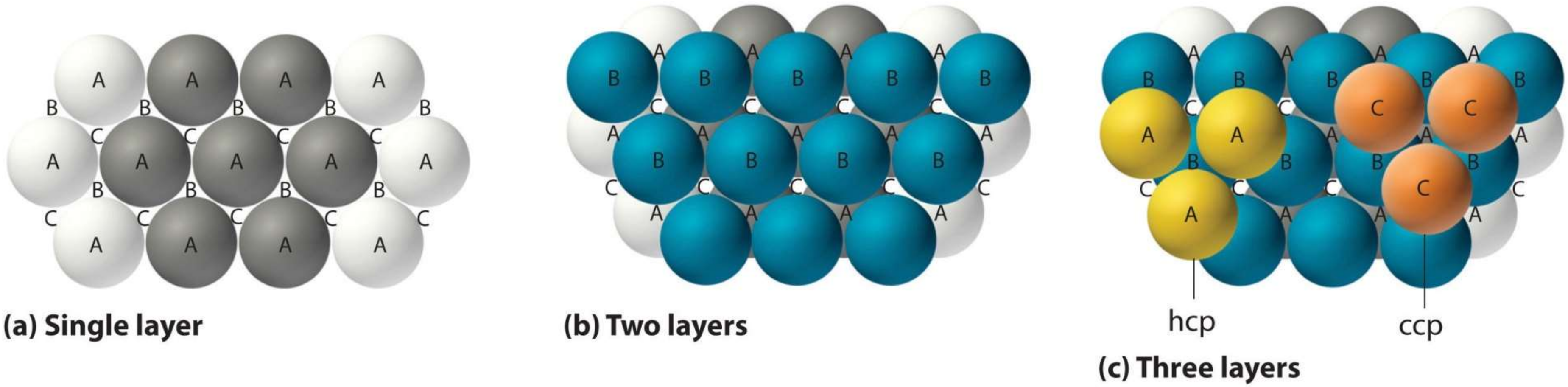
- **close packing**

TABLE 10.10 Summary of the Four Kinds of Packing for Spheres

Structure	Stacking Pattern	Coordination Number	Space Used (%)	Unit Cell
Simple cubic	*a-a-a-a-*	6	52	Primitive cubic
Body-centered cubic	*a-b-a-b-*	8	68	Body-centered cubic
Hexagonal closest-packed	*a-b-a-b-*	12	74	(Noncubic)
Cubic closest-packed	*a-b-c-a-b-c-*	12	74	Face-centered cubic



Particles /Cell	**1**	**2**	**4**	**6**

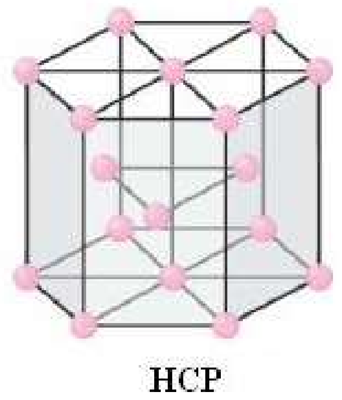


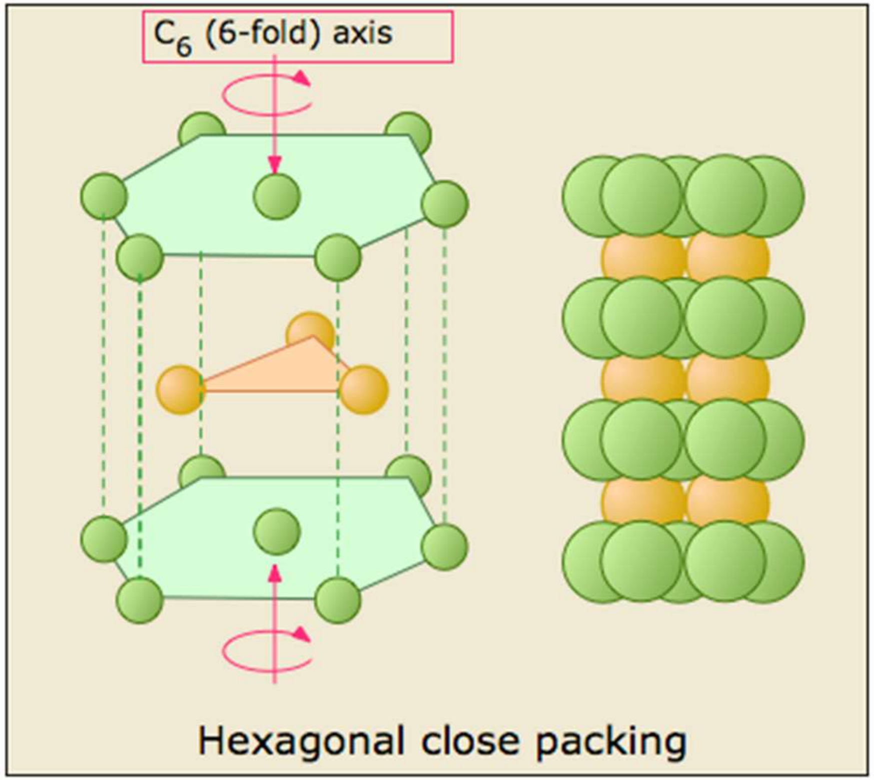
There are now two ways for placing the third layer:

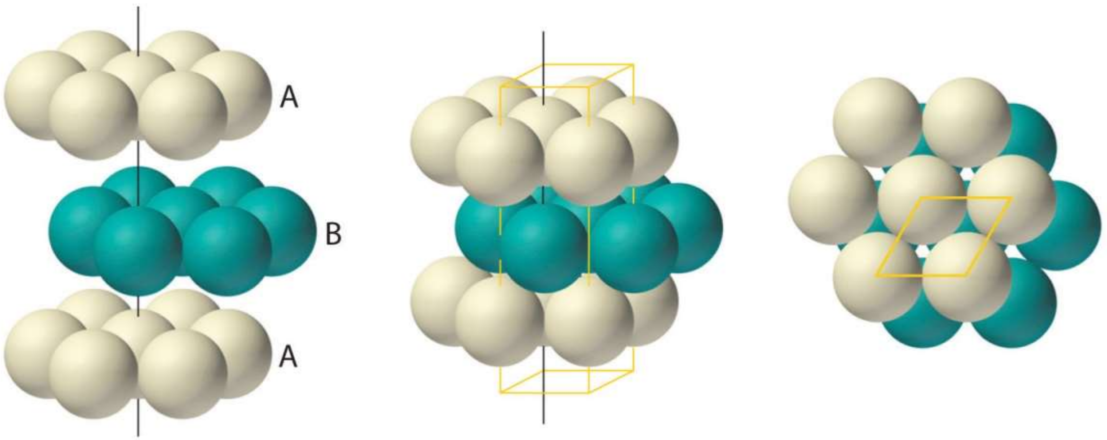
- If the atoms/ions in the third layer sit directly above the spheres in the first layer to form a ABABABA….pattern, the arrangement has a hexagonal unit cell, or is said to be hexagonal close packed (hcp).

- If the atoms/ions in the third layer are placed in C sites to form a ABCABC….pattern, the arrangement is called cubic close packing (ccp) or face centered cubic packing (fcc).

Hexagonal close packing (hcp) a ABABABA….pattern





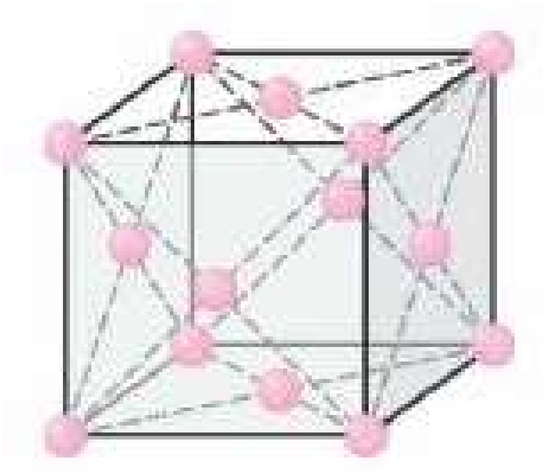


(a) Hexagonal close-packed (hcp)

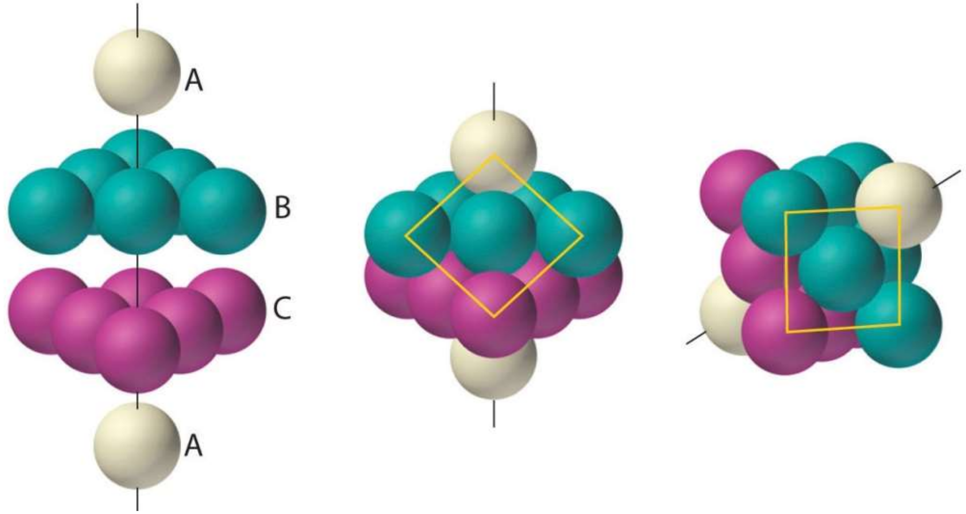
Coordination number
12

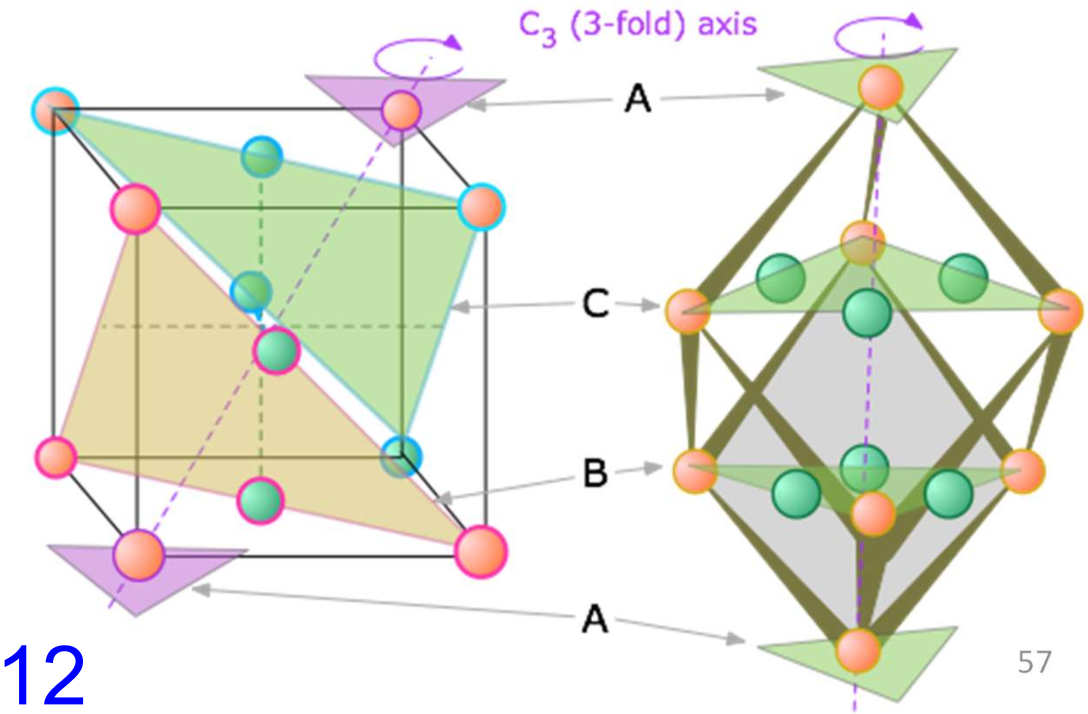
Face centered cubic close packing (fcc)

a ABCABC….pattern



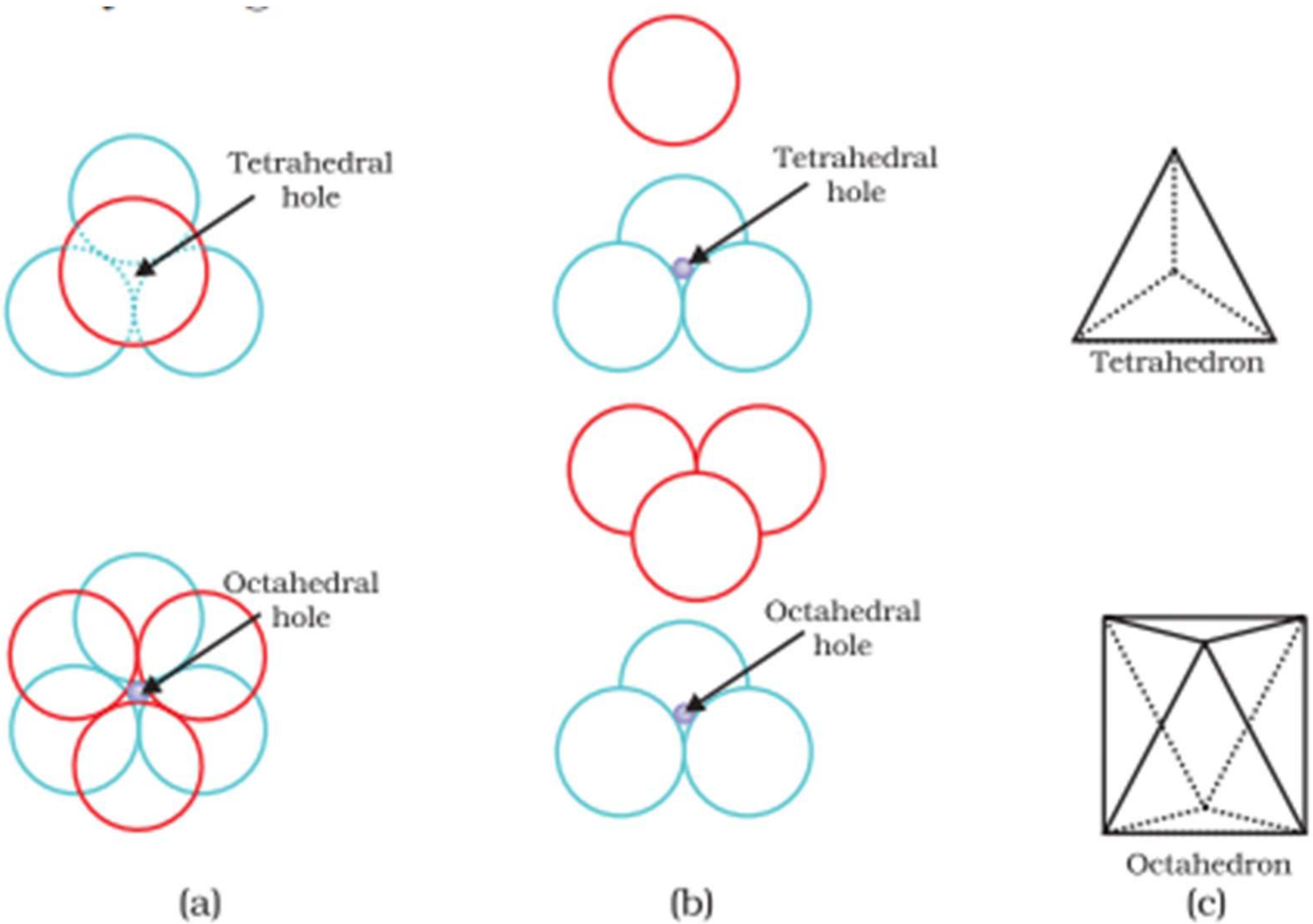
FCC

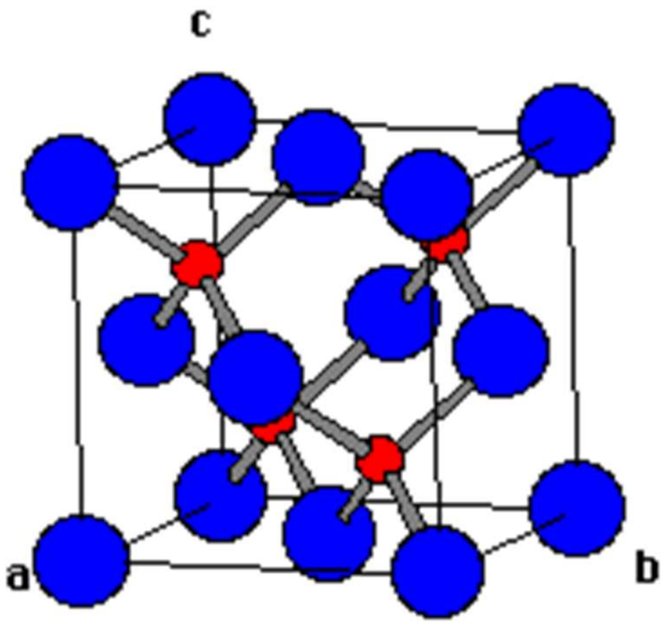




Coordination number 12

- **Interstitial voids**





Blue is S^{2-}

Red is Zn^{2+}

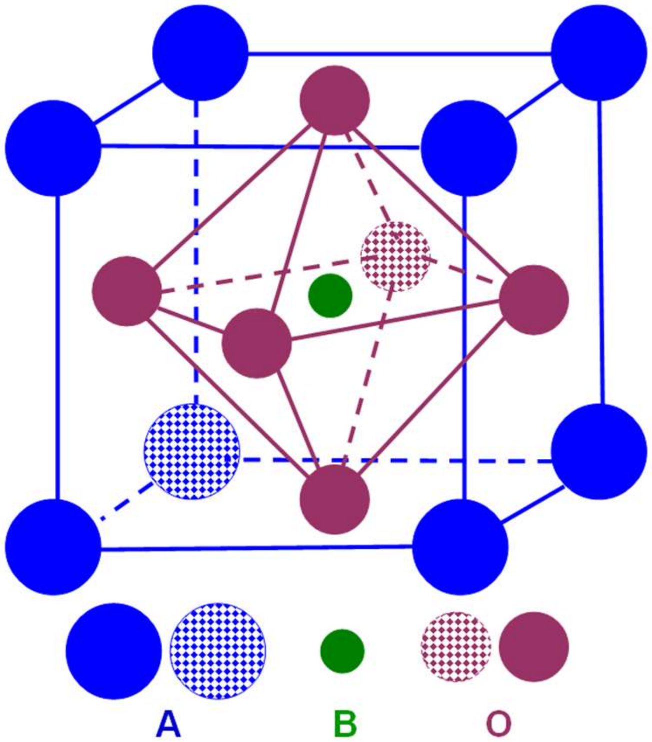
S^{2-}

8 on the corners: $8 \times {}^{1}/_{8} = 1$
6 on the faces: $6 \times {}^{1}/_{2} = 3$
Total: 4

Zn^{2+}

4 inside: $4 \times 1 = 4$
Total: 4

So the formula is ZnS

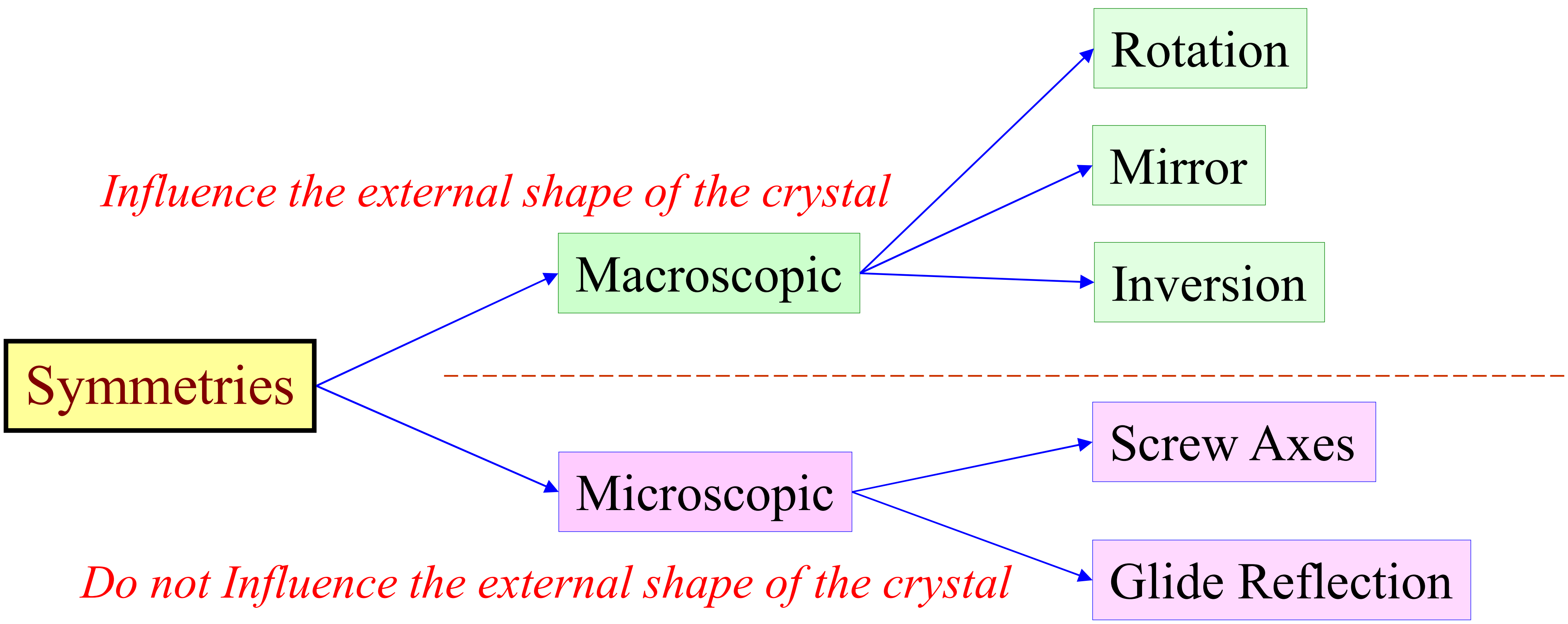


Perovskite ABO_3

Symmetry operation

A Symmetry operation is an operation that can be performed either physically or imaginatively that results in no change in the appearance of an object.

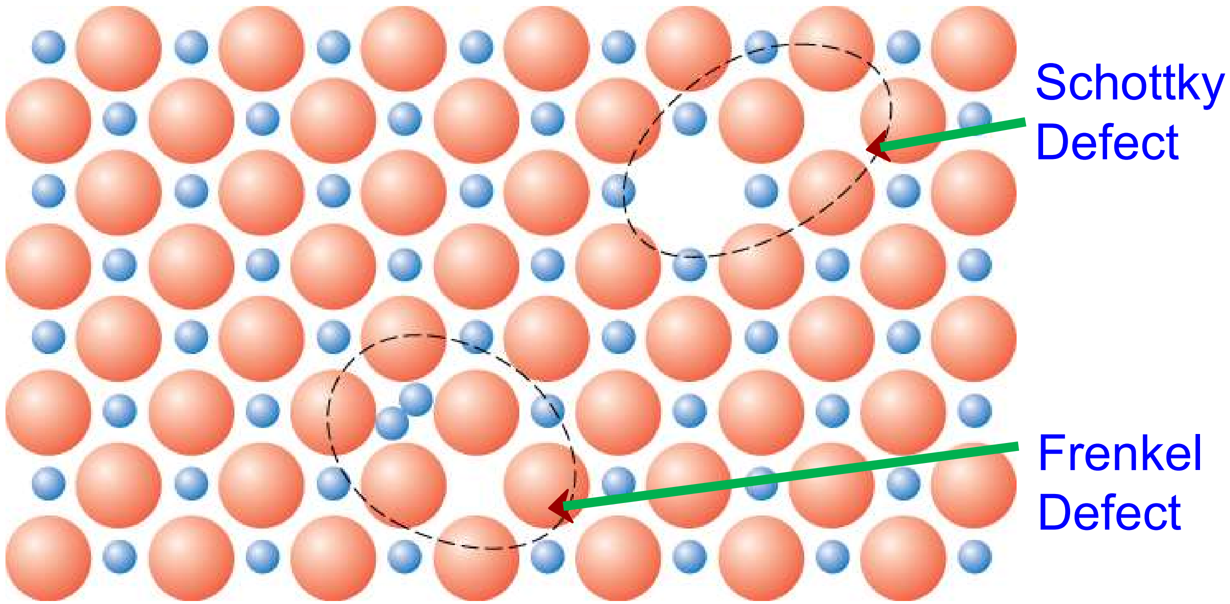
In crystallography, a crystallographic point group is a set of symmetry operations, like rotations or reflections, that leave a central point fixed while moving other directions and faces of the crystal to the positions of features of the same kind. Total 32 point groups.



The combination of all symmetry elements including the translation symmetry elements yields only 230 combination. These combinations are called “**space groups**”.

4. Defects

- Point defects



Schottky----vacancy正负粒子尺寸接近，配位数高 (+-size close)
Frenkel-----interstitial正负粒子尺寸差异大，配位数低
(+-big difference in size)

- Line defects--- dislocation

Types of dislocations: edge, screw, mixed.

- The strength of a material with no dislocations is 20-100 times greater than the strength of a material with a high dislocation density.
- So, materials with no dislocations may be very strong, but they cannot be deformed.
- The dislocations weaken a material, but make plastic deformation possible.
- The ability of a metal to deform depends on the ability of dislocations to move.
- Restricting dislocation motion makes the material stronger

1. Why metals could be plastically deformed?

The existence of dislocations

2. Why the plastic deformation properties could be changed to a very large degree by forging without changing the chemical composition?

3. Why plastic deformation occurs at stresses that are much smaller than the theoretical strength of perfect crystals?

Dislocations allow deformation at much lower stress than in a perfect crystal because slip does not require all bonds across the slip line to break simultaneously, but only small fraction of the bonds are broken at any given time.

- Planar defects--- grain boundaries

- fine-grained materials are usually much stronger than coarse-grained ones.

1. Powder Metallurgy

Ore

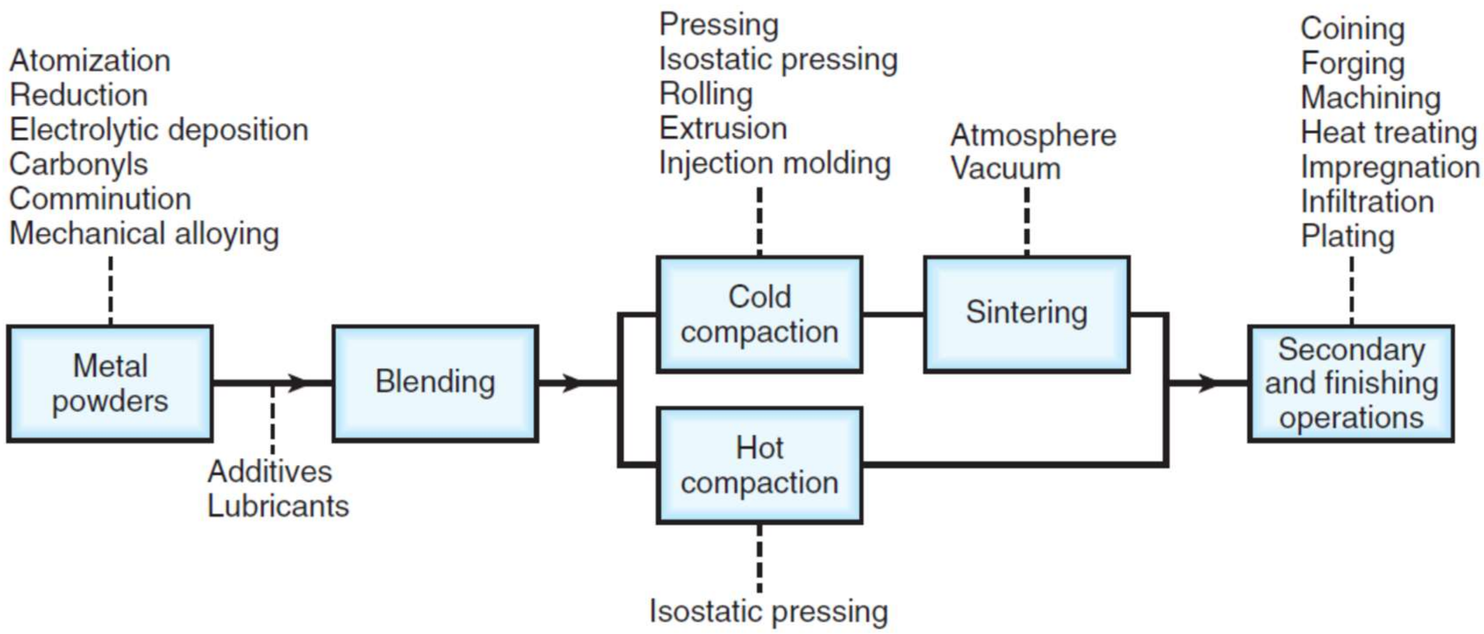
The mineral deposit contains an economically recoverable amount of a metal.

Gangue

The waste material of the rock formation.

The yield for metal extracting is quite low; ores typically possess less than 1% of the desired metal.

Metals



Powder-metallurgy process consists of:

A. **Powder production** **Atomization**

B. **Blending**

C. **Compaction** **CIP and HIP**

D. **Sintering**

E. **Finishing operations**

Advantages of powder metallurgy:

1. Lack of machining eliminates scrap losses.

2. Facile alloying of metals.

3. *In situ* heat treatment is useful for increasing the wear resistance of the finished material.

4. Facile control over porosity and density of the green and sintered material.

5. Fabrication of complex/unique shapes which would be impractical/impossible with other metalworking processes.

6. Rapid solidification process extends solubility limits, often resulting in novel phases.

Limitations of powder metallurgy:

- <u>Costs of powder production</u>
- Limitations on the shapes and features which can be generated e.g. the process cannot produce re-entrant angles by fixed die pressing or radial holes in vertically pressed cylinders
- <u>Pyrophoric risks</u>

  -- The extremely large exposed surface area of powders relative to the bulk results in rapid oxidation upon exposure to air;

  -- there is enhanced internal friction among the individual micron- or nano-sized individual particulates comprising the powder.
- <u>Potential workforce health problems from atmospheric contamination of the workplace</u>

2. The difference between microstructure and phase

What is meant by the terms ‘microstructure’ and ‘phase’

Microstructure:

- A small scale structure of a material.
- Multiple components and phases.
- Can be observed by microscope.
- Strongly influence material properties, such as strength, toughness, ductility, hardness, and corrosion resistance.

Phase:

A ‘phase’ is taken to be any part of a material with a distinct crystal structure and/or chemical composition. Different phases in a material are separated from one another by distinct boundaries.

2. Fe-C alloy and the phase diagram

Four phases in Fe-C alloys

- **α-ferrite - solid solution of C in BCC Fe**
- **γ-austenite - solid solution of C in FCC Fe**
- **δ-ferrite solid solution of C in BCC Fe**
- **Fe_3C (iron carbide or cementite)**

Temperature increase

Cementite

- Hard and brittle
- Cementite is actually a metastable phase, with graphite representing the most stable form of carbon at equilibrium. However, it is difficult to obtain stable graphite nuclei in steels (with < 2%C) since the Fe/Fe_3C metastable equilibrium is preferentially formed.

Pure Fe VS Fe-C alloy

- **Strength (Hardness)** ↑
- **Ductility** ↓
- **Electrical conductivity** ↓
- **Density** ↓

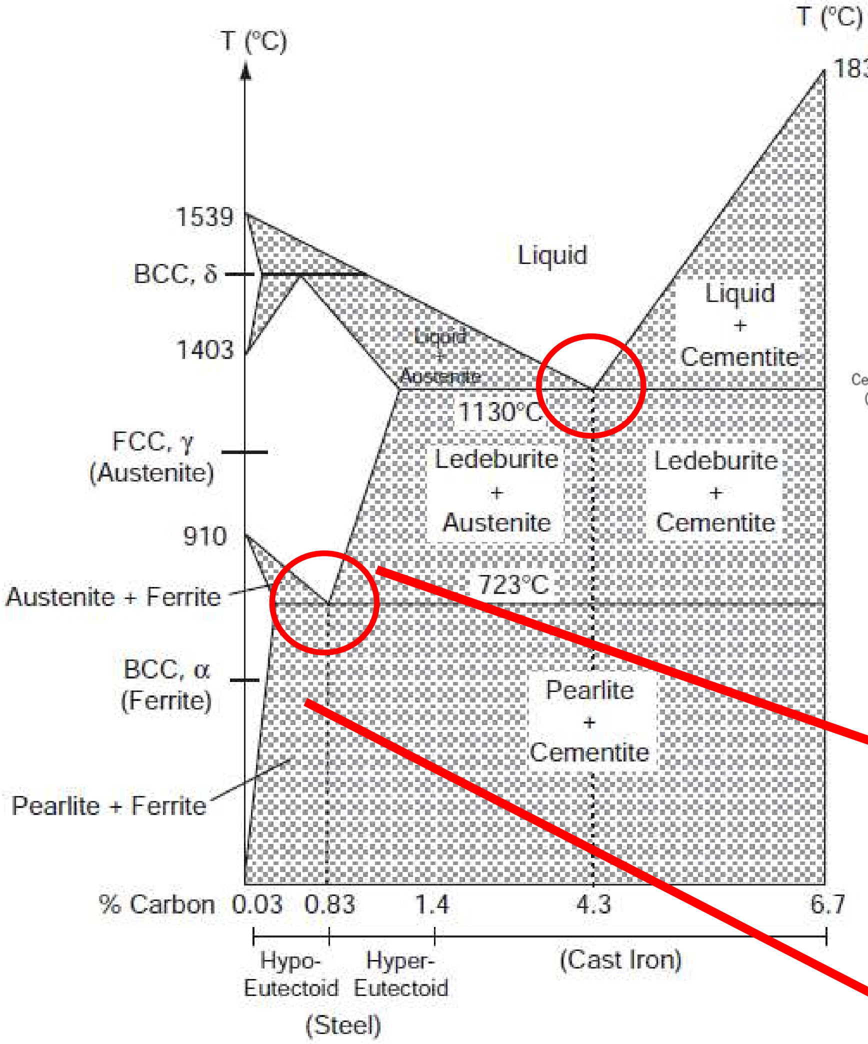
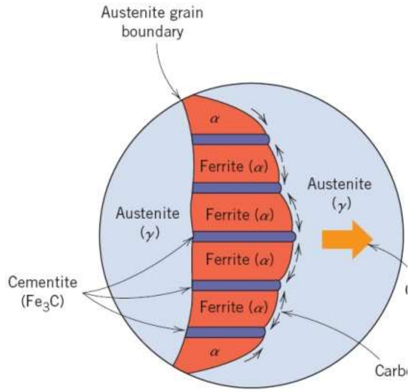


Figure 3.10. The equilibrium phase diagram for the iron–carbon system.



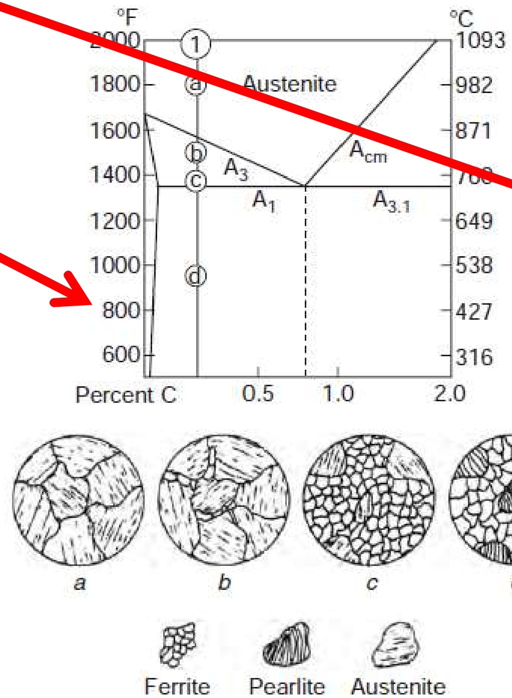
Microstructure: pearlite

a lamellar structure of two phases: α-ferrite and cementite

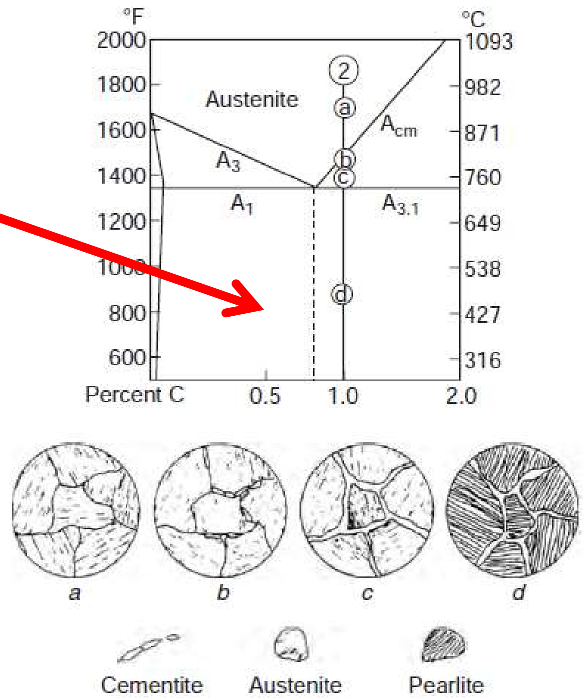
Eutectoid point: 0.83 wt%; for steel

Hypoeutectoid steel < 0.83 wt% <Hypereutectoid steel

Pearlite+ferrite



Pearlite+cementite

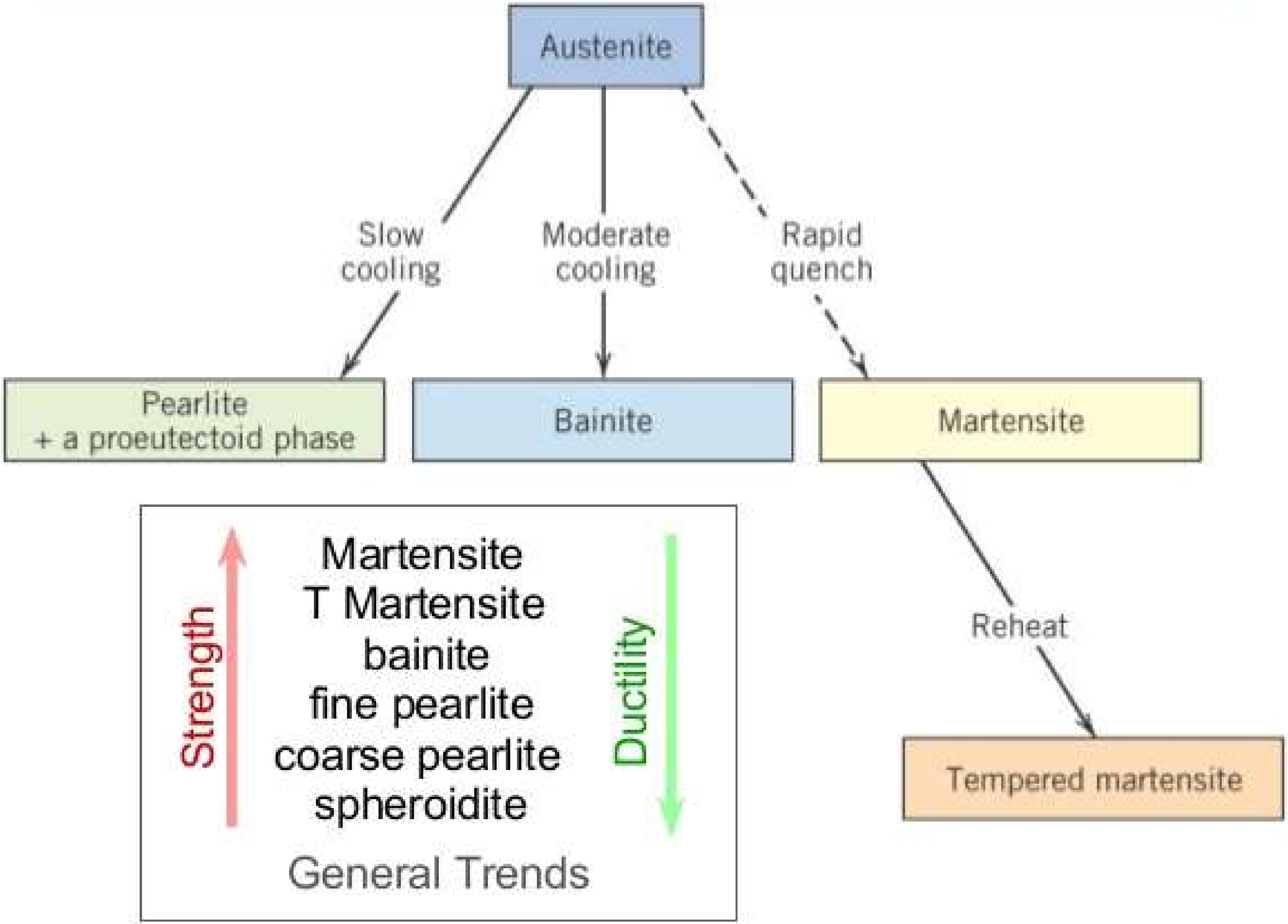


3. Hardening methods

Hardening strategies:

- Strain hardening,
- grain size hardening,
- Precipitation alloying, Transformation hardening
- Solute hardening,
- Surface hardening.

Possible Transformations



Concepts

- **Semiconductors**

Semiconductors possess electrical conductivity intermediate between conductors such as metals and insulators such as ceramics.

- **Intrinsic semiconductor**

Intrinsic semiconductors contain the same numbers of free bonding electrons, and “holes” created from the migration of electrons from the valence to conduction bands.

- **Extrinsic semiconductor**

For extrinsic semiconductors, the Fermi level corresponds to a level slightly above or below conduction or valence bands, depending on the dopants introduced into the lattice – either an excess of electrons or holes

Semiconducting Materials

- **n-type semiconductor**

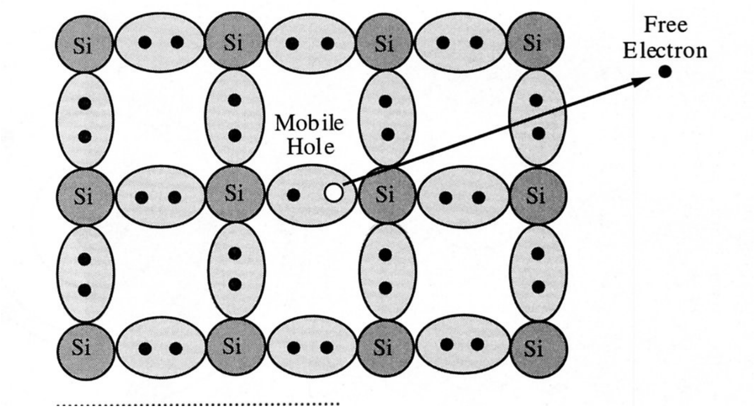
When a small amount of pentavalent impurity (such as N, P) is added to a pure semiconductor crystal during crystal growth, the resulting crystal is called the N-type semiconductor. An N-type semiconductor material has an excess of electrons.

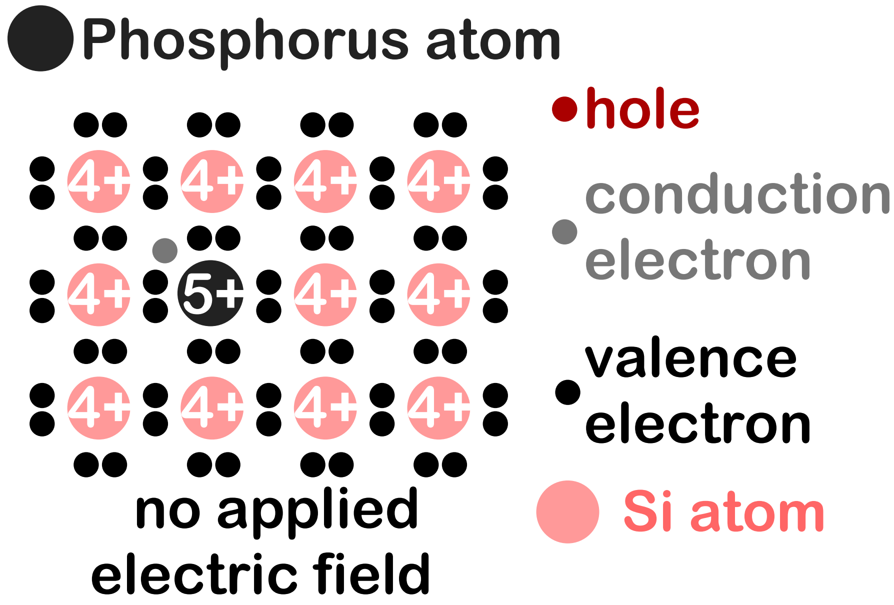
- **p-type semiconductor**

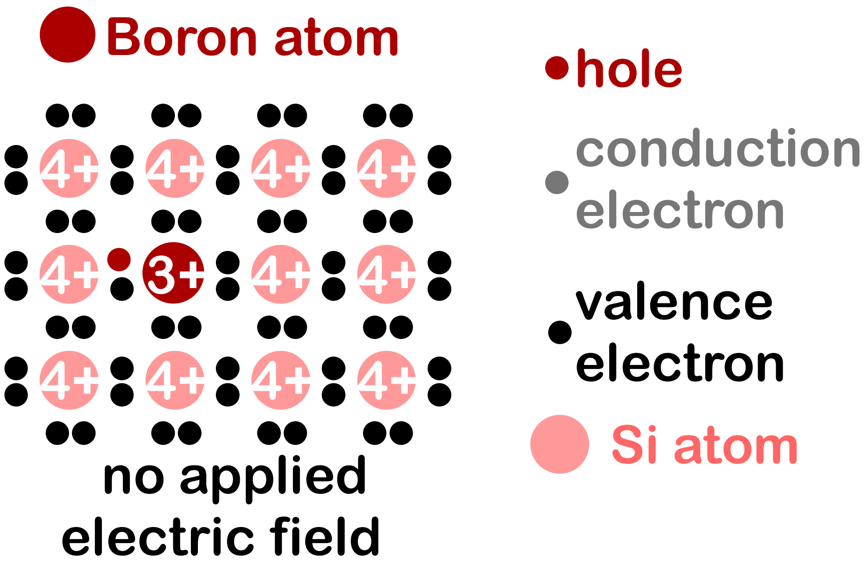
When a small amount of trivalent impurity (such as B, Al) is added to a pure semiconductor crystal during crystal growth, the resulting crystal is called a P-type semiconductor. In a p-type semiconductor material there is a shortage of electrons, i.e. there are 'holes' in the crystal lattice.

- **Fermi level**

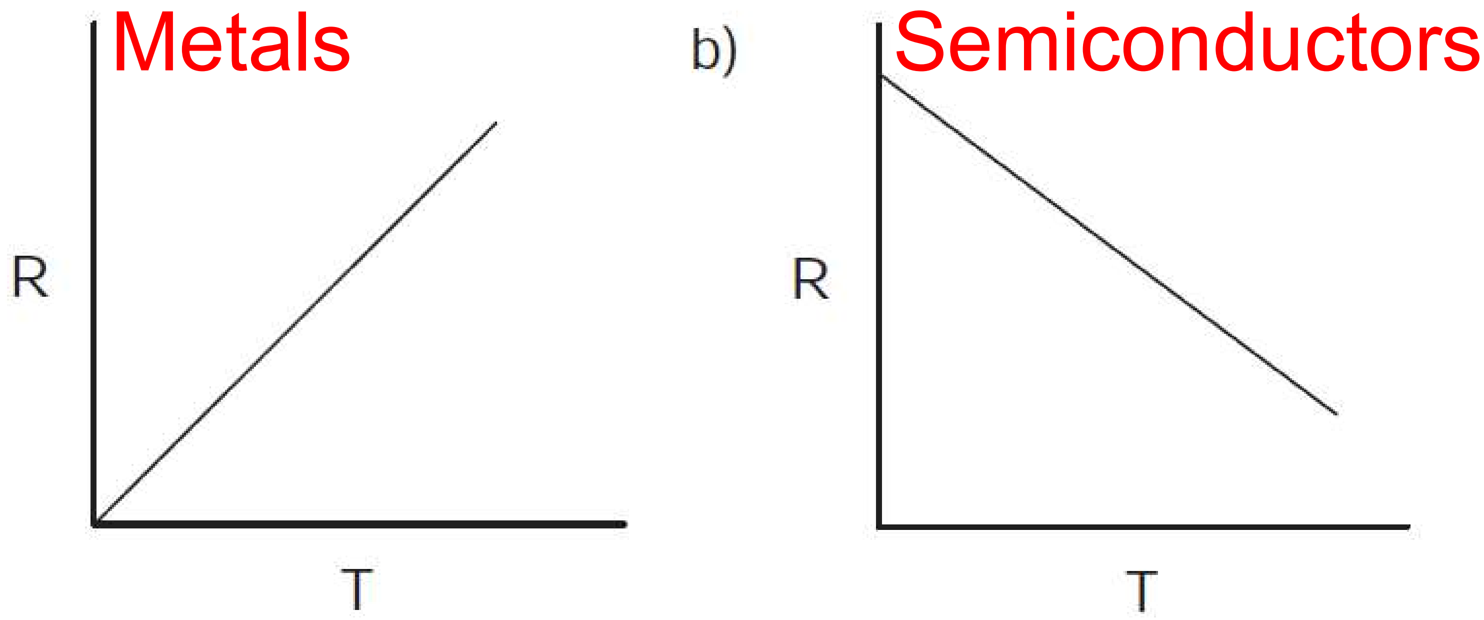
Fermi level is the highest energy state occupied by electrons in a material at absolute zero temperature.


Si
Free
Electron
Mobile
Hole


Phosphorus atom
hole
conduction
electron
valence
electron
Si atom
4+
5+
no applied
electric field


Boron atom
hole
conduction
electron
valence
electron
Si atom
4+
3+
no applied
electric field

Temperature dependent conductivity



As the temperature is increased:

- Metals show a decreased conductivity.
- Semiconductors show an increased conductivity.

The thermal motion of metal atoms causes less efficient electron mobility through the lattice, whereas a temperature increase causes the bandgap to narrow for semiconductors, resulting in more effective electrical conductivity.

Silicon Wafer

Silicon is the second most abundant element in the earth's crust, next to oxygen.

Silicon wafer is fabricated from sea sand.

Metallurgical-grade silicon (MG-Si) >98%.
Electronic-grade silicon (EG-Si) 99.999999999%.

Semiconducting devices

P-n junction

P-n junction diode

A PN Junction Diode is one of the simplest semiconductor devices around, and which has the characteristic of <u>passing current in only one direction</u> only.

MOSFET transistors:

Metal Oxide Semiconductor Field Effect Transistors

nMOSFET: n-channel with p-type substrate

pMOSFET: p-channel with n-type substrate

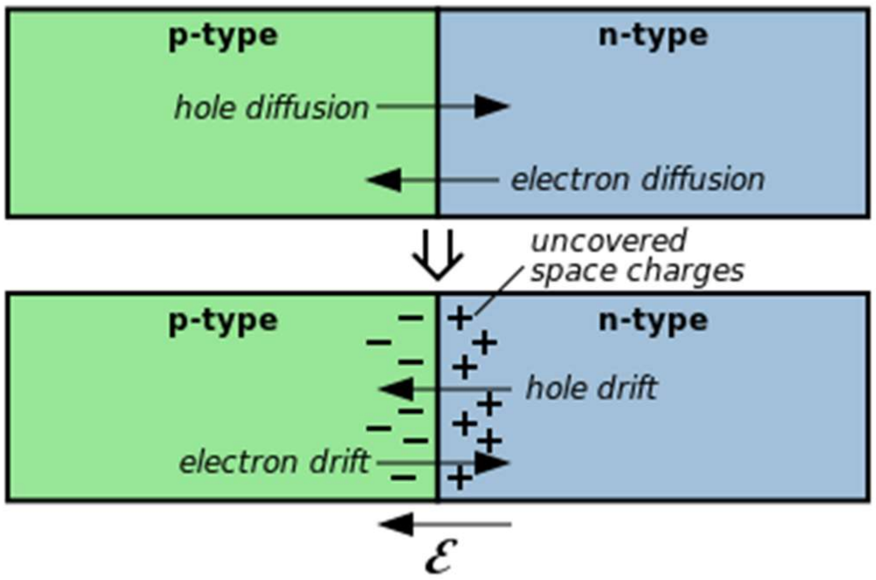
Complementary Metal–Oxide–Semiconductor (CMOS)

CMOS technology involves the complimentary operation of interconnected nMOSFET and pMOSFET pairs.

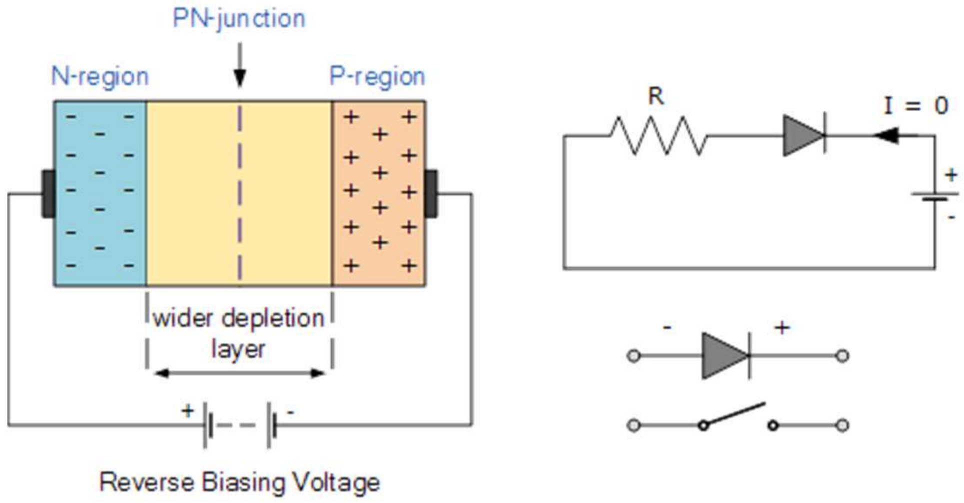
Semiconducting Materials

p-n junction diode

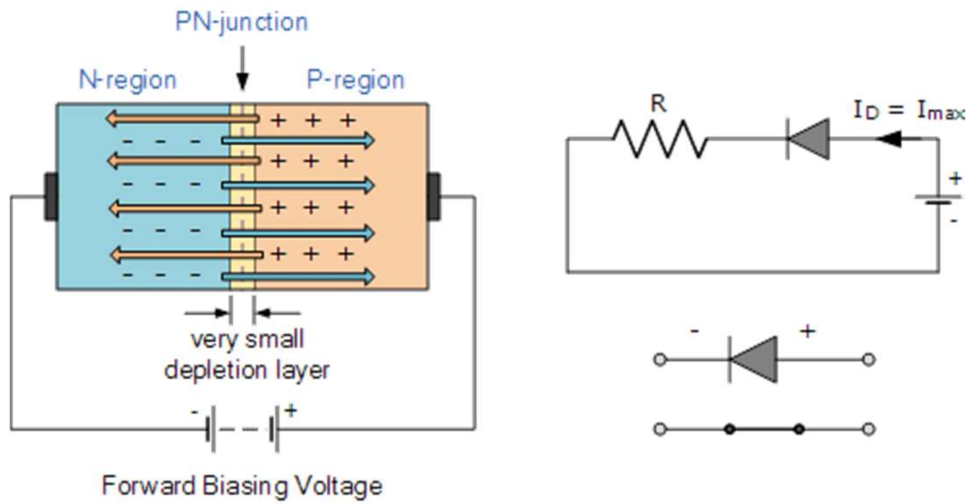
Depleted region

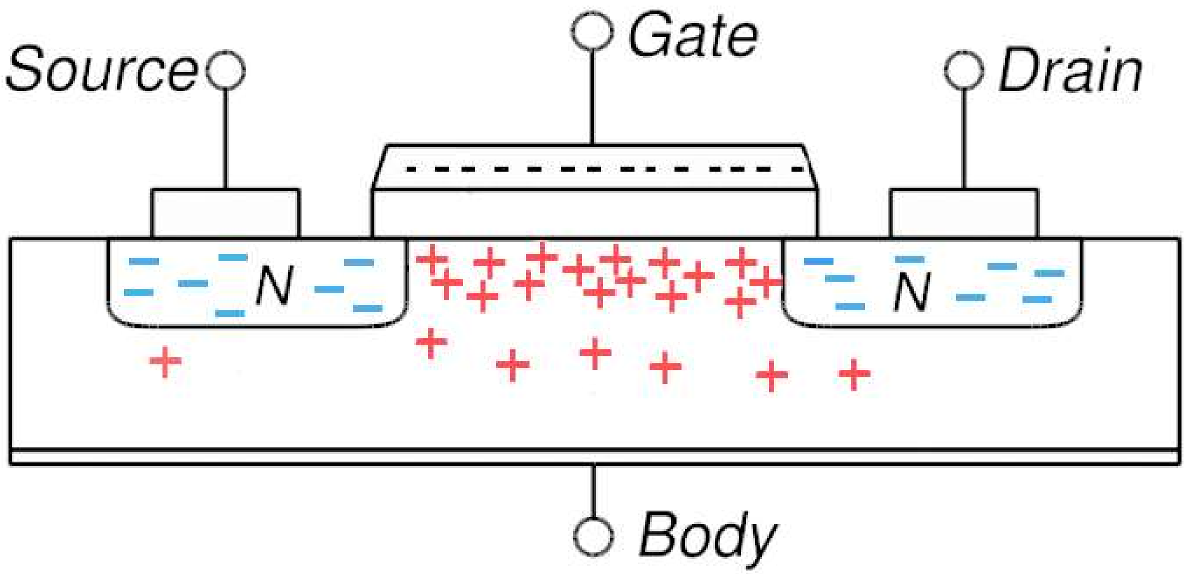


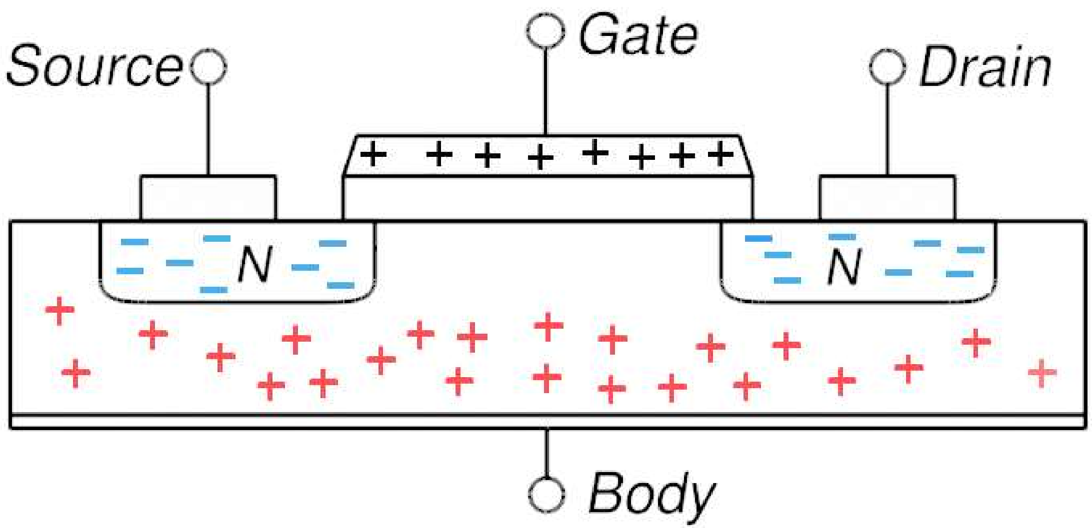
Reverse Bias



Forward Bias




Source
Gate
Drain
N
N
Body


Source
Gate
Drain
N
N
Body

Integrated circuit fabrication

- Majority of ICs utilize a Si(100) substrate **WHY?**

That is, since the Si(100) surface is the least dense, there will be a relatively low number of coordinatively unsaturated (*i.e.*, dangling) Si bonds at the surface.

- The fabrication of a CMOS IC involves a repeating sequence of film deposition, patterning, and dopant implantation procedures.

Semiconducting Materials

$$I_{\mathrm{DS}} = \frac{W}{2Lt_{\mathrm{ox}}}\mu C(V_{\mathrm{GS}} - V_{\mathrm{T}})^2$$

I_{DS}: the drain current

↑↑

W: the channel width
μ: the channel carrier mobility
C: the capacitance density of the gate oxide
V_{GS}: the gate voltage
V_T: the threshold voltage

↑↓

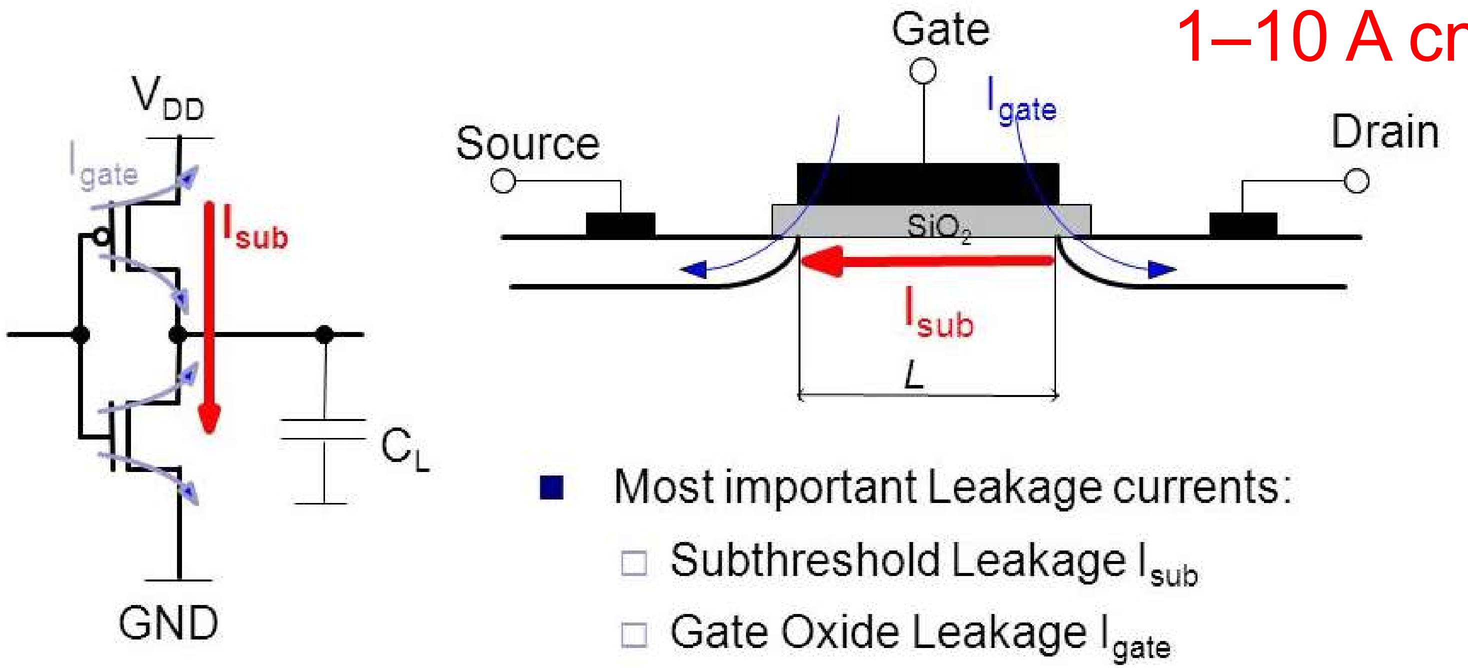
L: the channel length
t_{ox}: the gate oxide thickness

The smaller the better?

A thin insulating layer is prone to electron tunneling between the gate and channel, leading to increased power consumption and heat buildup.

Leakage current density

Current transistors: 1–10 A cm^{-2}



Empirical and theoretical studies have shown that the practical limit for dielectric effectiveness of the gate SiO_2 layer is *ca*. 0.7–1 nm.

To solve the Gate Leakage problems

SiO_2 gate oxide ⟹ Insulating film High-κ dielectrics

$\kappa = 3.82$ $\kappa > 3.82$

$$C_{vol} \propto \frac{\kappa}{t^2}$$

$$\kappa \uparrow \rightarrow t \uparrow$$

C_{vol}: the capacitance per unit volume;

κ: the dielectric constant;

t; the thickness of the dielectric medium.

1. What is a semiconductor? What is an extrinsic semiconductor? Describe the difference between P-type and N-type semiconductor materials.
2. At what temperature are there no electrons in the conduction band of a semiconductor?

   0 K

3. As one electron is promoted from the valence band to the conduction band, a_______ is formed in the valence band. hole

4. Group ____ elements are used as dopants to produce n-type semiconductors, because they have __________________________than the original Group IV material.

   V; one more electron

5. Group ____ elements are used as dopants to produce p-type semiconductors, because they have __________________________than the original Group IV material.

   III; one less electron

6. A diode contains both ______________and ____________regions. P-type; n-type

7. For current to flow through a diode, the positive terminal of the power supply must be connected to the ____-type material. p

8. Majority of integrated circuits utilize a Si (100) substrate, Why?